

KernelAscent: RSI benchmark — figure atlas

September 18, 2026

Auto-generated figure atlas (regenerate: `python3 scripts/make_paper_figures.py`). Pick the figures that matter; each has a `label` for cross-referencing.

1 Leaderboards

Model	tier	rounds	$L - F$
starcoder2-15b	large	6	0.232
deepseek-coder-6.7b-base	mid	3	0.154
starcoder2-15b	large	5	0.071
deepseek-coder-1.3b-base	small	12	0.039
deepseek-coder-1.3b-base	small	8	0.038
deepseek-coder-1.3b-base	small	4	0.038
deepseek-coder-1.3b-base	small	24	0.001
Qwen2.5-Coder-3B	small	8	0.000
Qwen2.5-Coder-14B	large	4	-0.000
Qwen2.5-Coder-1.5B	small	3	0.000
Qwen2.5-Coder-1.5B	small	3	0.000
Qwen2.5-Coder-1.5B	small	12	0.000

Table 1: Task 5 open self-play: author co-evolution $L - F$ leaderboard.

Model	Q_0	Q_g	Δ vs base
us.openai.gpt-6-astra	0.298	0.854	0.556
us.anthropic.claude-sonnet-5	0.493	0.894	0.401
mistral.mistral-large-3-675b	0.359	0.669	0.310
Nova-Pro	0.401	0.490	0.089
us.openai.gpt-5.6-sol	0.813	0.889	0.075
Kimi-K2.5	0.752	0.813	0.062
Claude-Opus-5	0.878	0.910	0.032
Fable-5.1	0.732	0.741	0.009
us.openai.gpt-5.6-terra	0.851	0.850	-0.001
Llama-3.3-70B	0.458	0.455	-0.004
deepseek.v3.2	0.847	0.614	-0.234

Table 2: Task 3 procedure-RSI: held-out Q gain vs. frozen procedure.

Size band	n	P(cross wall)	P(RSI)	mean drift
<2B	26	0.54	0.04	0.024
2-8B	17	0.94	0.41	0.299
$\geq 9B$	7	0.71	0.29	0.396

Table 3: Scale $\rightarrow$ RSI gates: correctness-wall crossing and compounding by size band.

2 Mechanistic interpretability

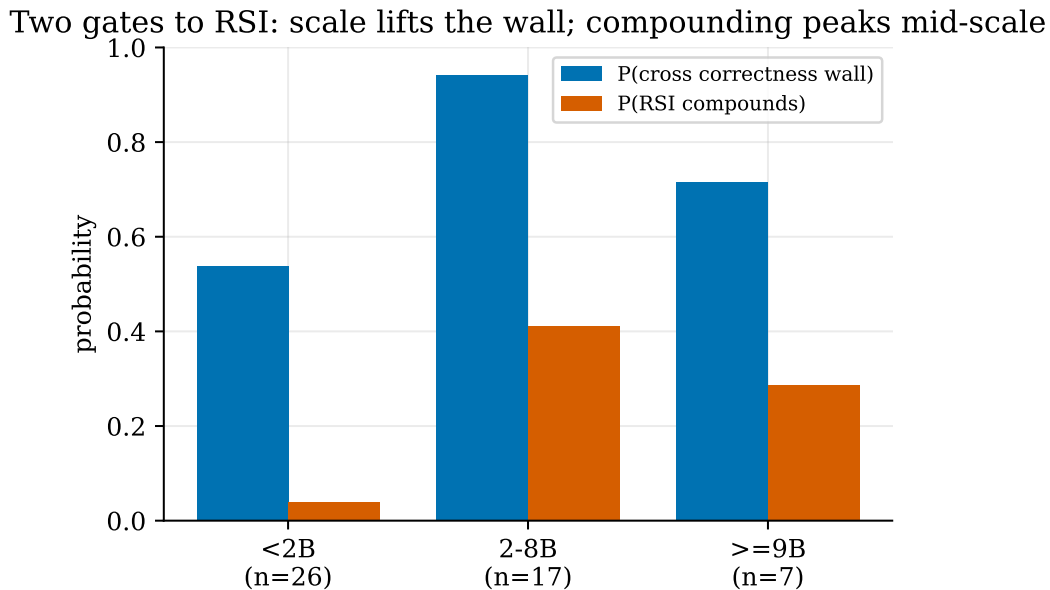


Figure 1: Scale $\rightarrow$ RSI. Below $\sim 2B$ the correctness wall is rarely crossed, so weight-RSI is impossible; crossing rises sharply at $2B+$, but compounding peaks at mid-scale ($2-8B$) and does not increase further $\geq 9B$ (headroom ceiling).

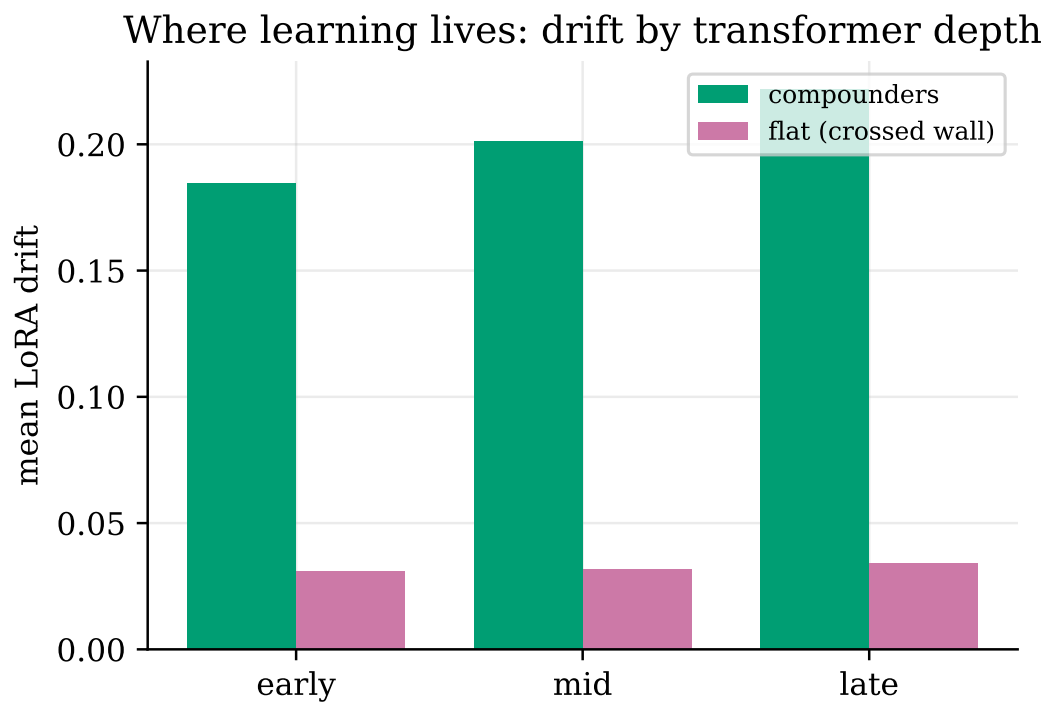


Figure 2: LoRA weight drift by transformer depth for compounds vs. flat (wall-crossing) runs. Compounds sustain markedly larger drift across all depths; flat runs saturate early.

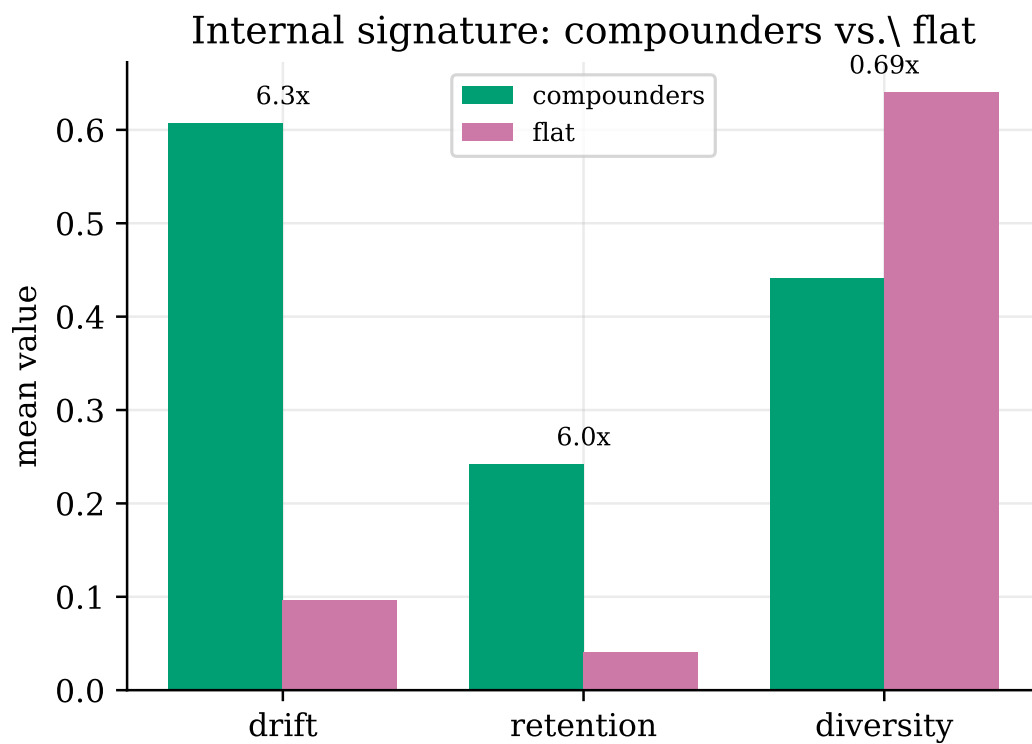


Figure 3: The internal signature that separates compounders from flat runs among wall-crossers: sustained drift and retention are $\sim 5\text{--}6\times$ higher in compounders, while generation diversity is *lower* (productive convergence, not diversity preservation).

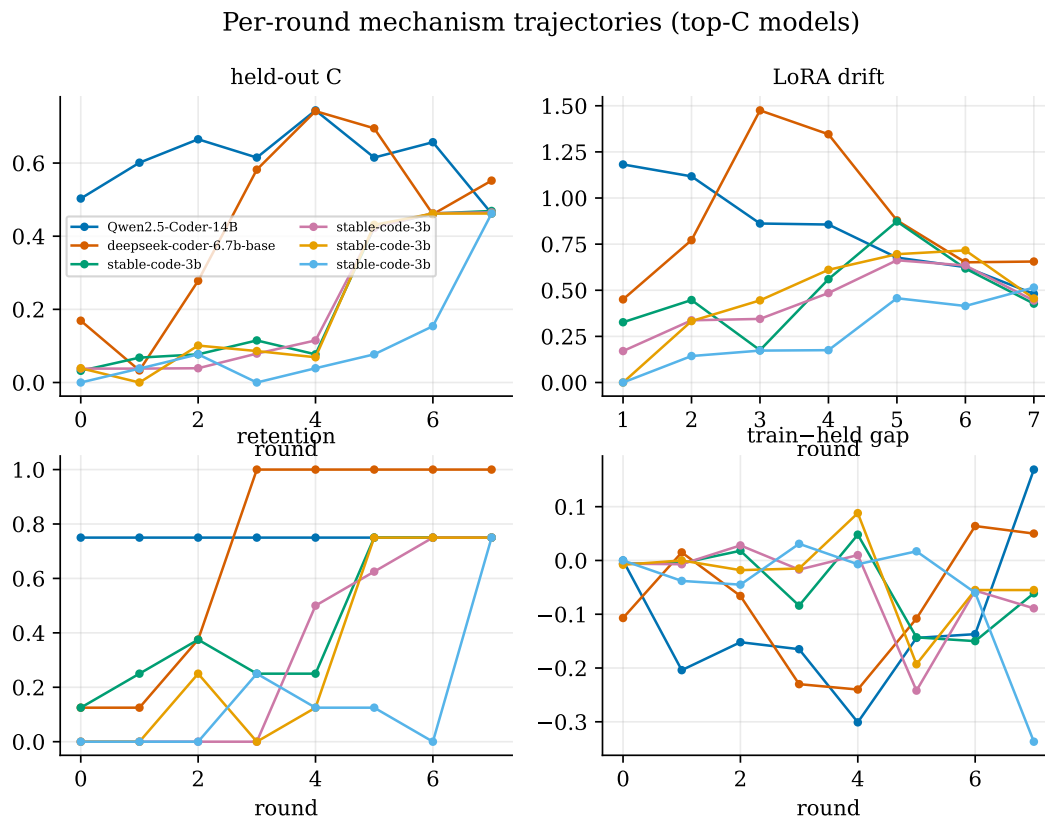


Figure 4: Per-round mechanism trajectories for the highest-capability probes: held-out C, total LoRA drift, retention, and the train–held transfer gap. Compounders show rising/held C with sustained drift and retention; flat runs plateau with drift saturation.

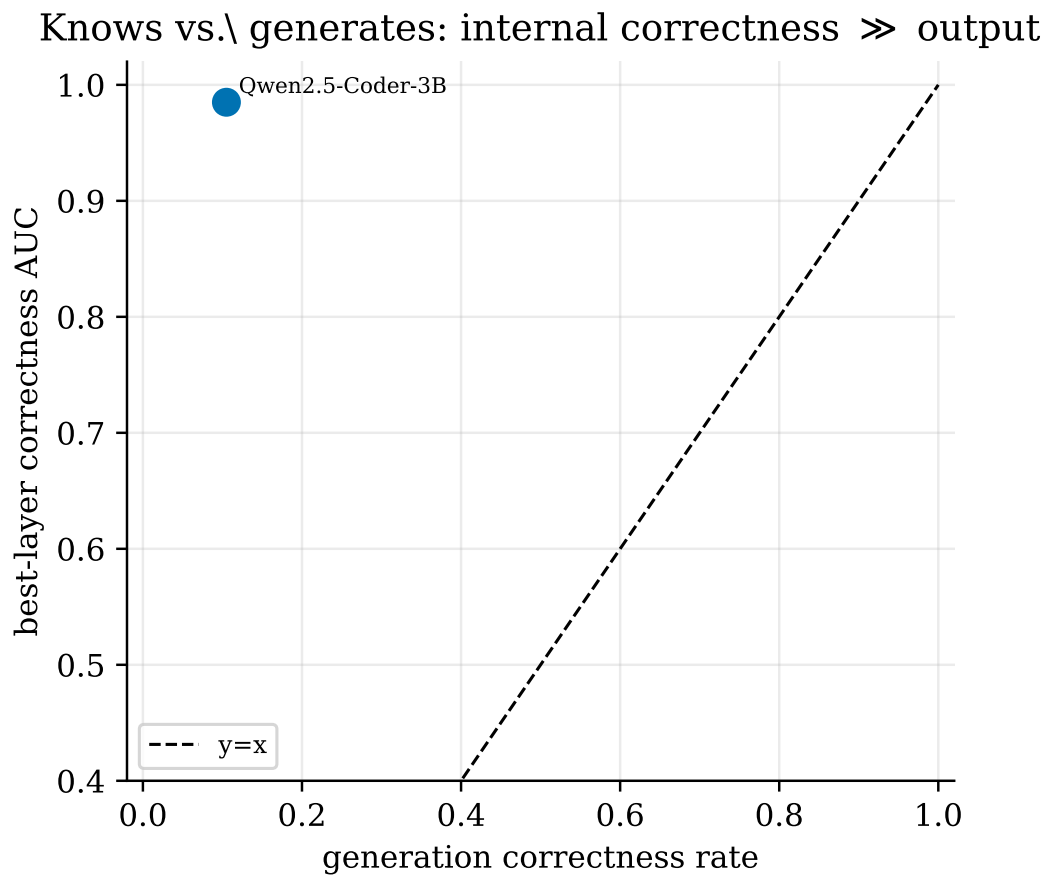


Figure 5: Internal correctness decodability (best-layer linear-probe AUC) vs. how often the model actually generates a correct kernel. Points far above $y = x$ mean the model *represents* correctness internally far better than it can *decode* it into a correct kernel: the RSI bottleneck is generation, not knowledge.

3 Task 5: self-play

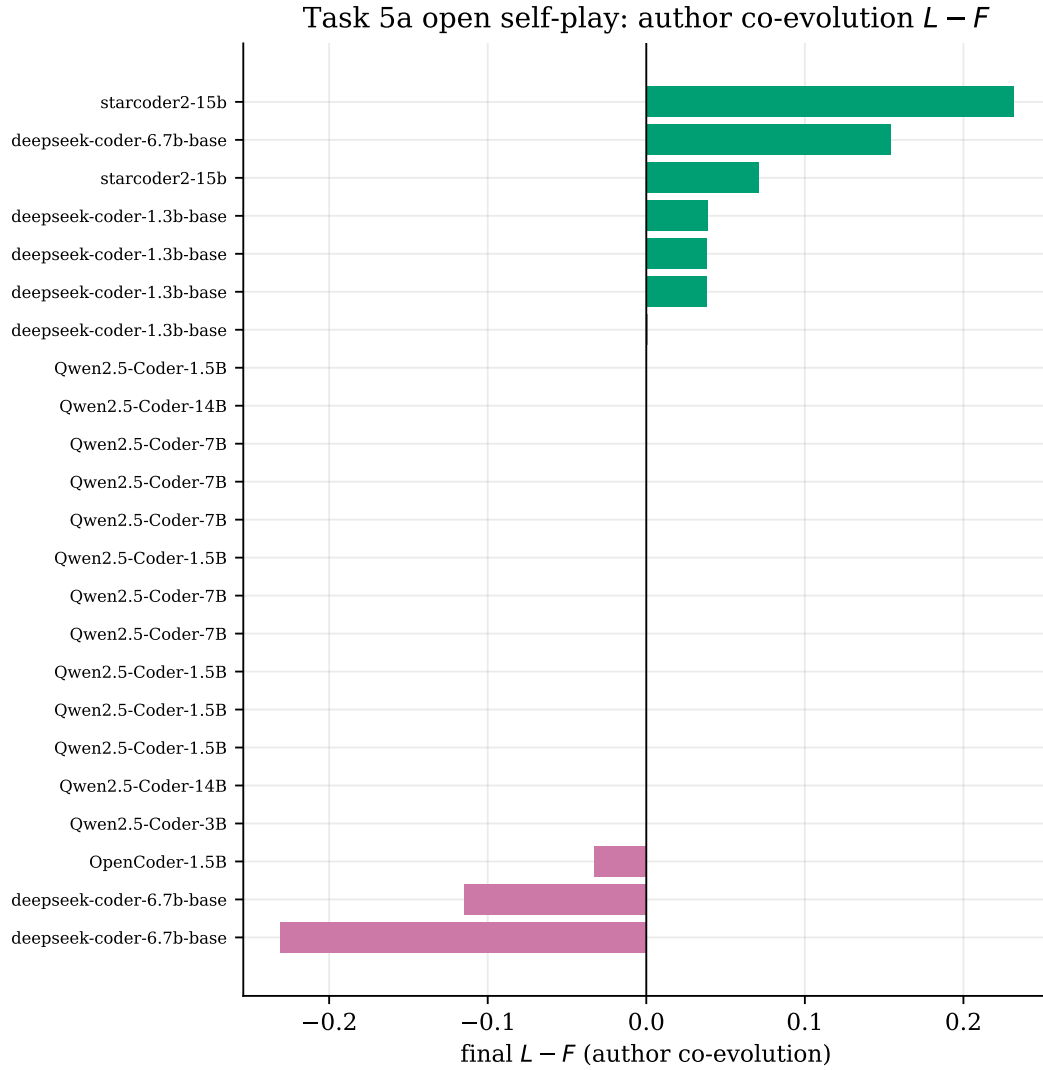


Figure 6: Final $L - F$ (live-author minus frozen-author, the self-referential signal) per open-weight model. > 0 means updating the author beyond a frozen-author escalating curriculum genuinely helps.

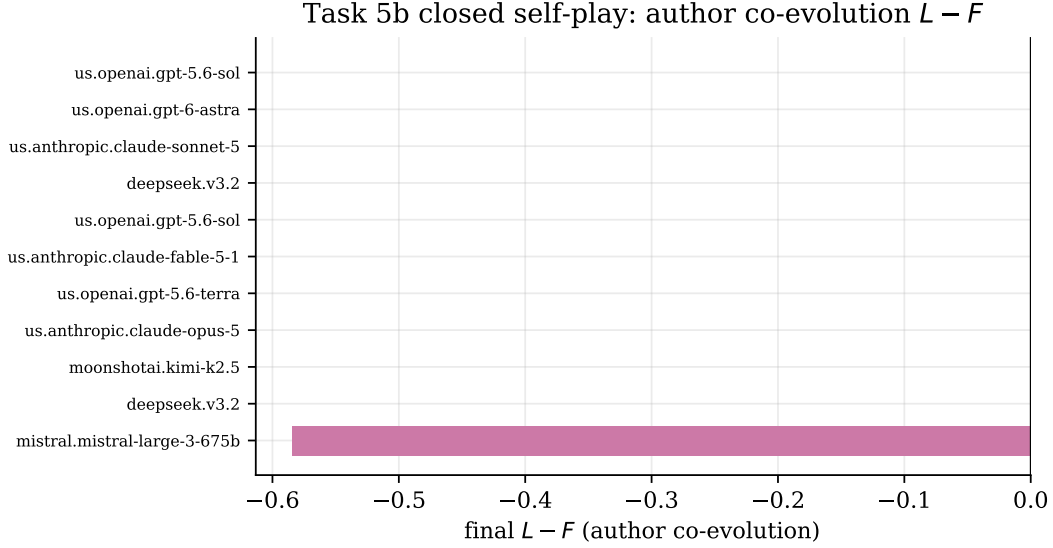


Figure 7: Final $L - F$ per closed/API model (procedure co-evolution channel). Most cluster near zero; early large spikes did not survive longer horizons.

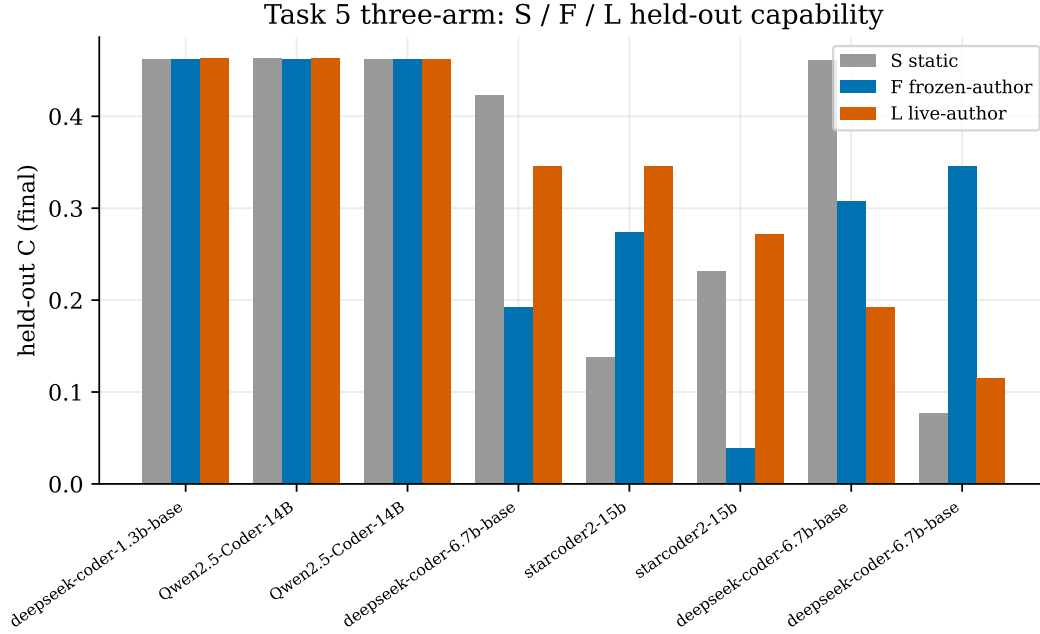


Figure 8: Three-arm self-play at equal budget on a fixed held-out ladder: STATIC (fixed frontier), FROZEN-AUTHOR (escalating frontier, frozen base author), LIVE-AUTHOR (co-evolving author). $F - S$ isolates the adaptive-curriculum benefit; $L - F$ isolates author co-evolution.

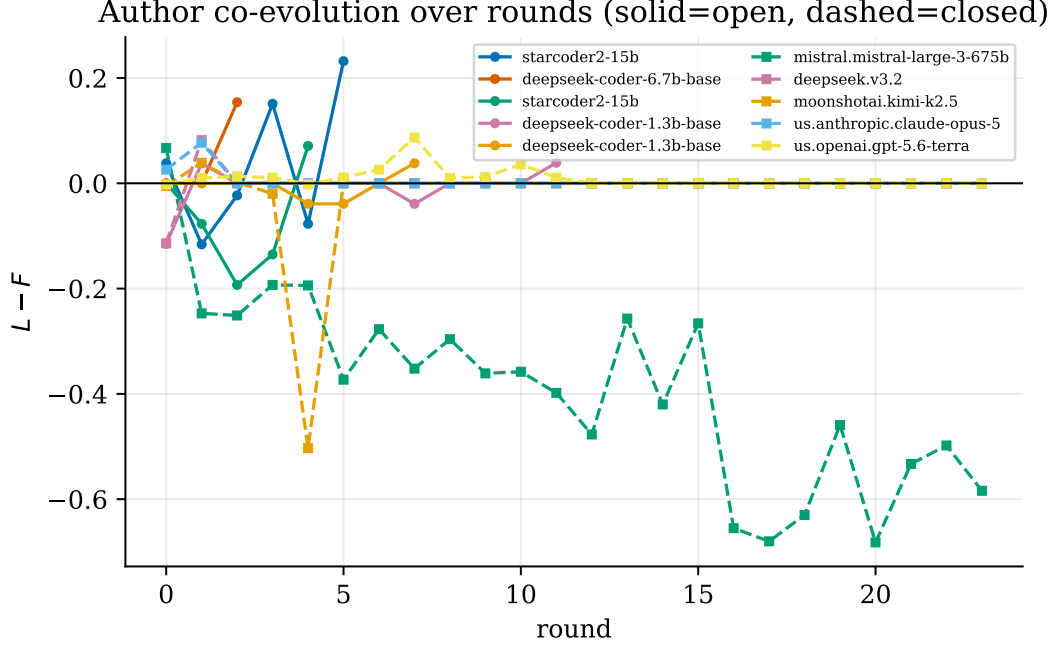


Figure 9: Per-round $L - F$ trajectories (solid = open-weight, dashed = closed/API). Long horizons reveal whether early positive spikes compound or regress to zero.

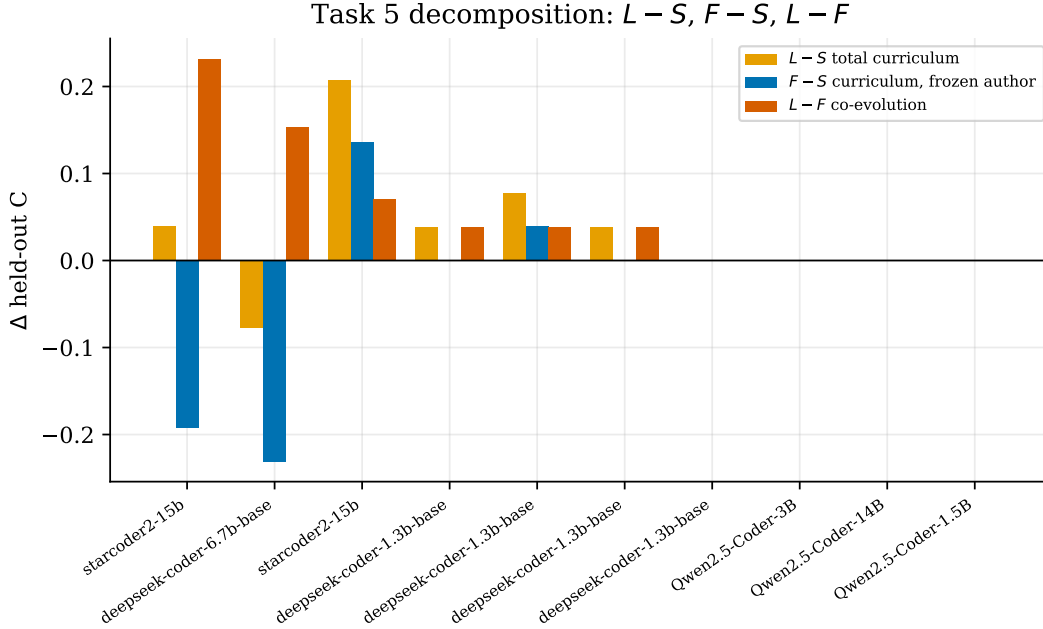


Figure 10: Decomposing the self-play benefit: $L - S$ (total adaptive-curriculum gain) = $(F - S)$ (curriculum with a frozen author) + $(L - F)$ (extra gain from co-evolving the author). Most models get their gain from $F - S$, not $L - F$.

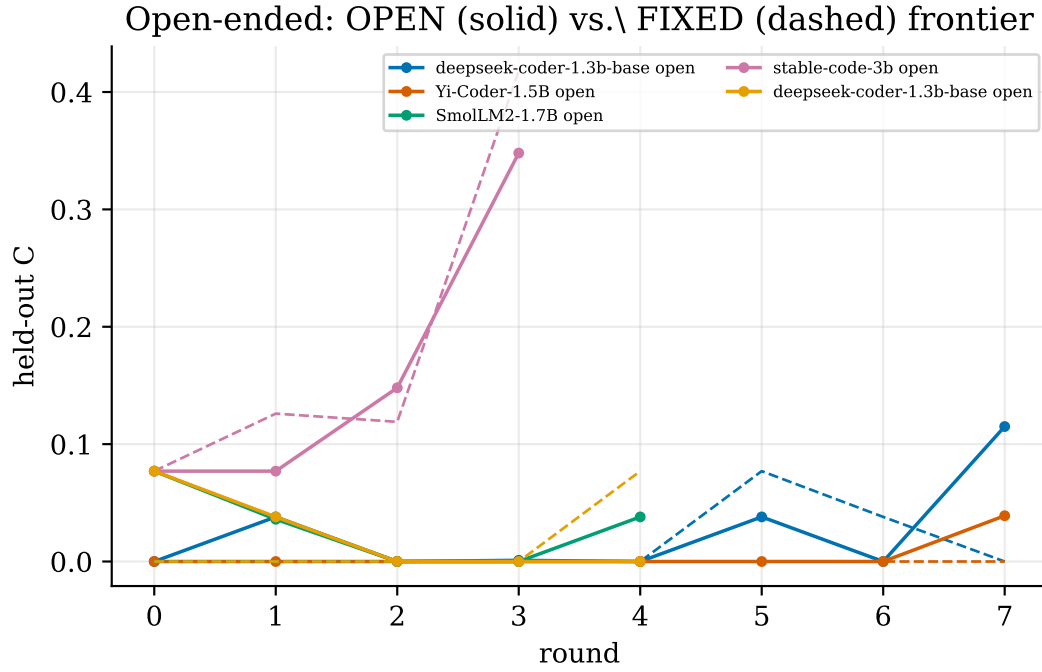


Figure 11: Open-ended RSI: a learner facing an OPEN escalating self-authored frontier (solid) vs. a FIXED static frontier (dashed) at equal budget. Overlap means open-endedness is a one-time upgrade, not compounding.

4 Task 4: closed→open

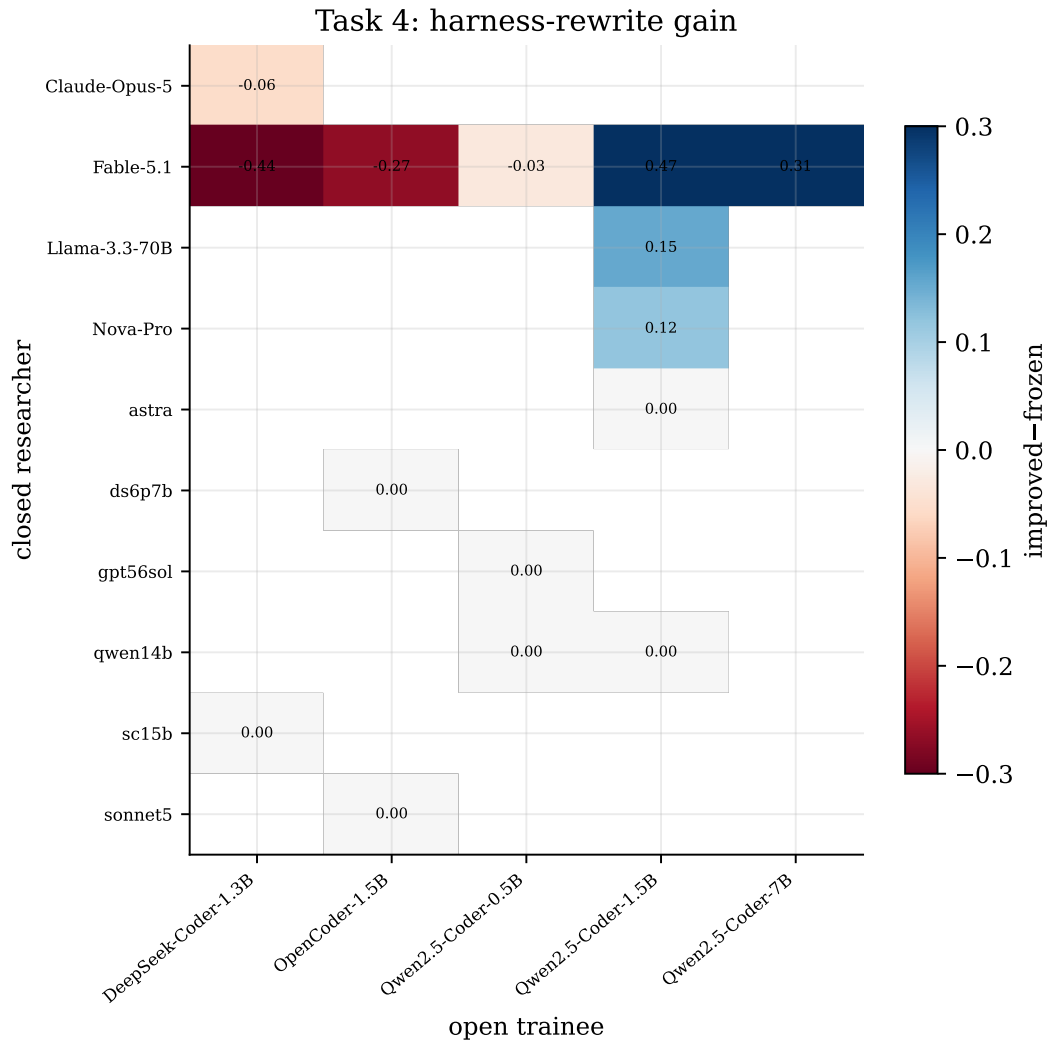


Figure 12: Task 4 (closed→open): a closed frontier researcher rewrites an open trainee’s training harness. Cell = improved – frozen held-out C on the trainee (positive = the rewritten harness helps).

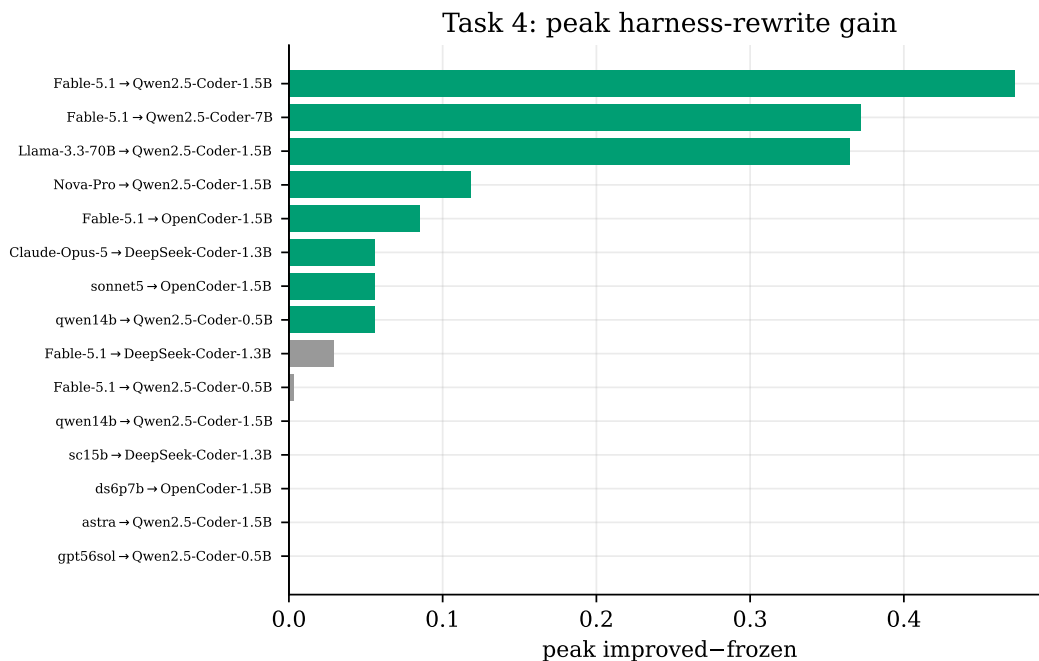


Figure 13: Peak improved-frozen over rounds for each researcher→trainee pair.

5 Task 3: procedure-RSI

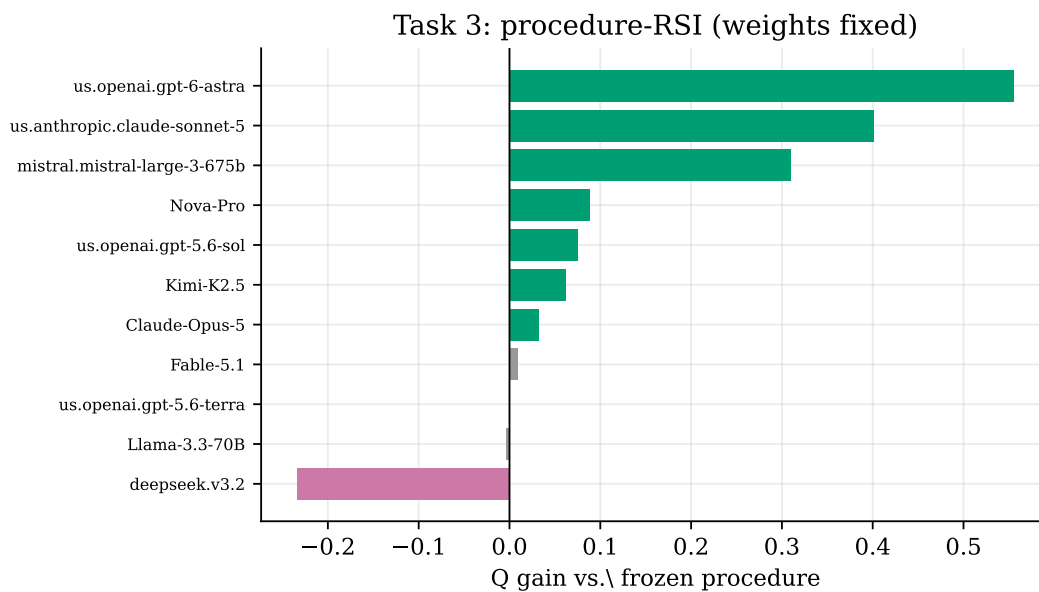


Figure 14: Task 3: a frozen-weight model rewrites its own strategy library + verified archive. Held-out Q gain vs. the frozen procedure; frontier models improve their own procedure substantially.

6 Comparators & weight-RSI

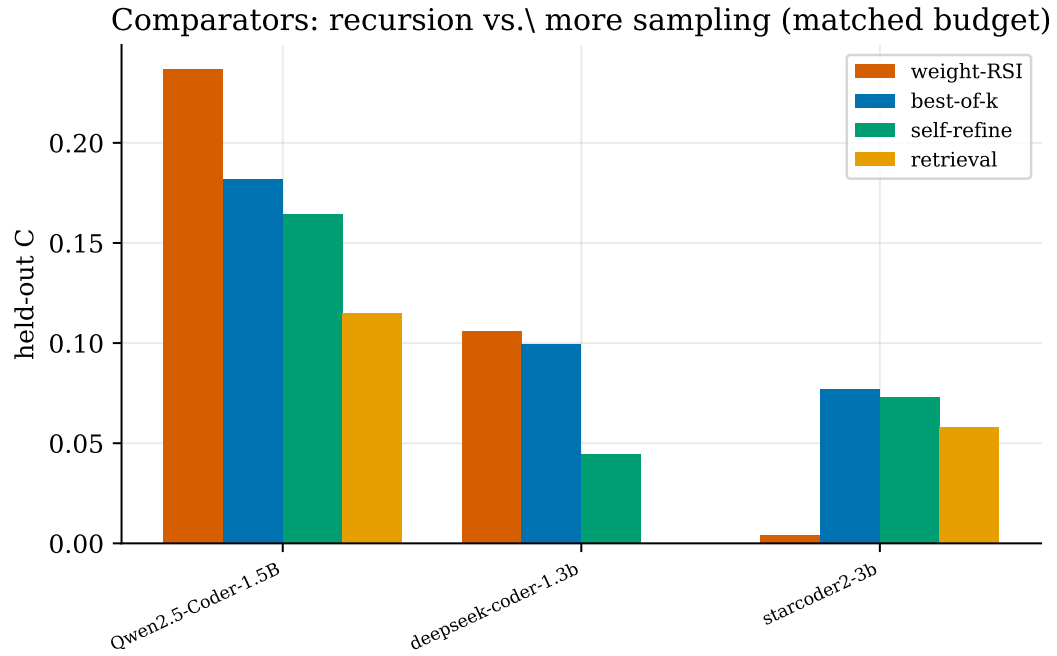


Figure 15: Is it really recursion? Weight-RSI vs. non-recursive controls (best-of-k, self-refine, retrieval) at matched generation budget. $\text{recursion_gain} = \text{weight-RSI} - \text{best comparator}$.

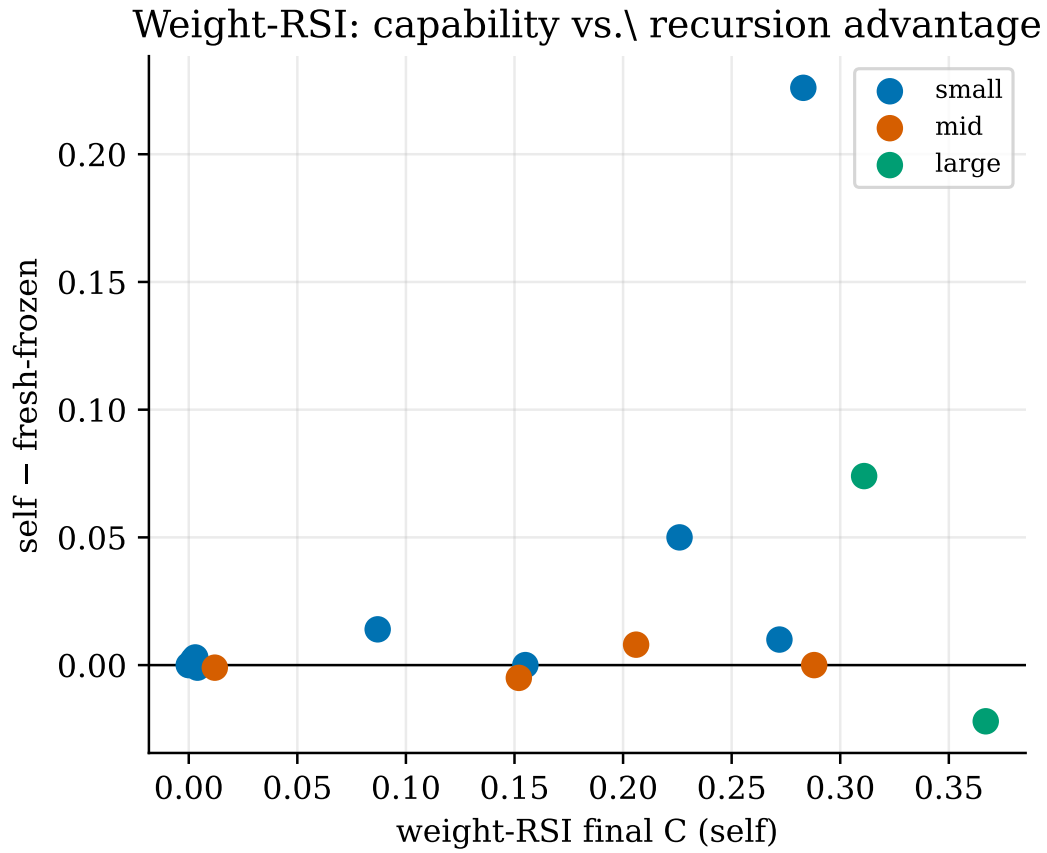


Figure 16: Weight-RSI: final self-trained held-out C vs. the self-minus-fresh-frozen advantage (positive = training on own kernels beats an equal-budget fresh baseline).

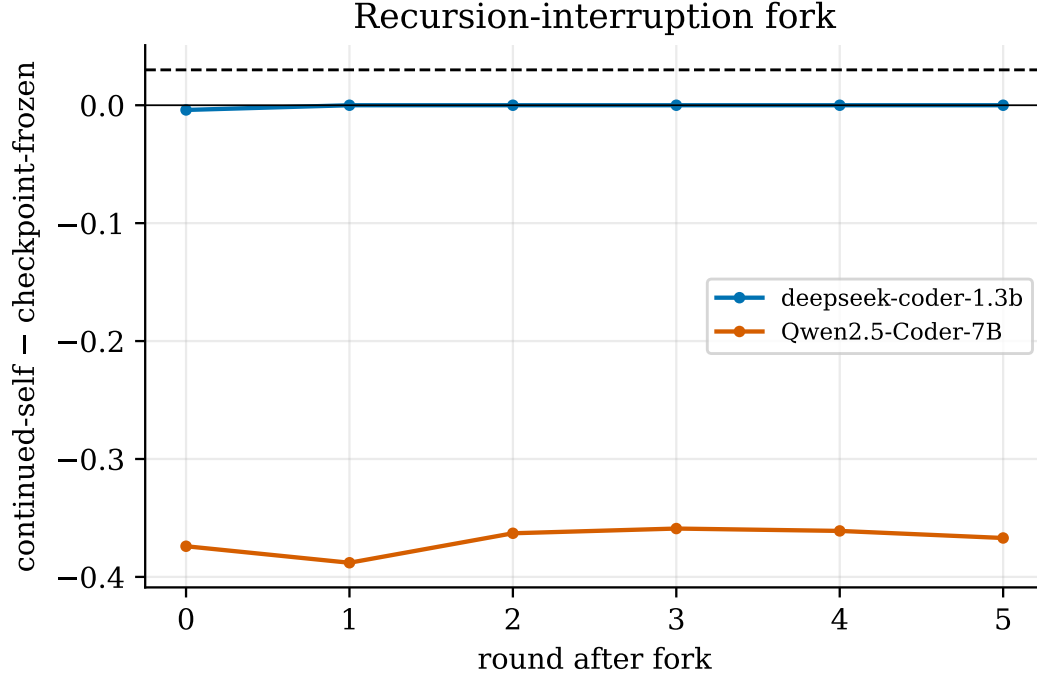


Figure 17: Recursion-interruption control: continued self-training vs. a checkpoint-frozen producer at equal budget. Sustained > 0 = genuine compounding rather than a one-time producer upgrade.

7 Flows & causality

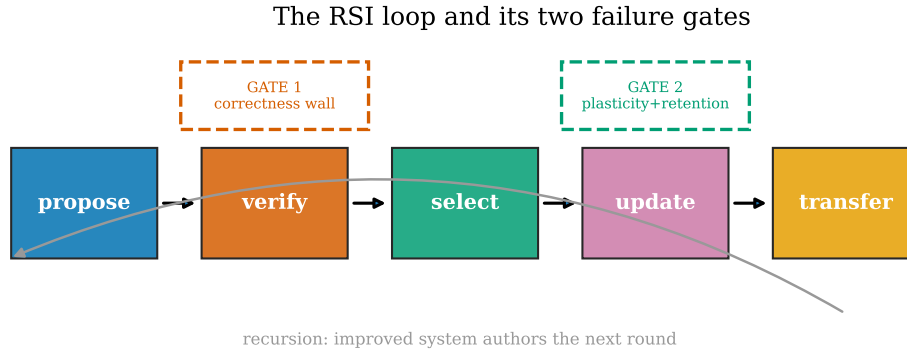


Figure 18: The recursive self-improvement loop (propose $\rightarrow$ verify $\rightarrow$ select $\rightarrow$ update $\rightarrow$ transfer) and the two gates our data identifies: Gate 1, the correctness wall (no verified kernel $\Rightarrow$ no gradient), and Gate 2, plasticity+retention (drift saturation / forgetting collapses compounding).

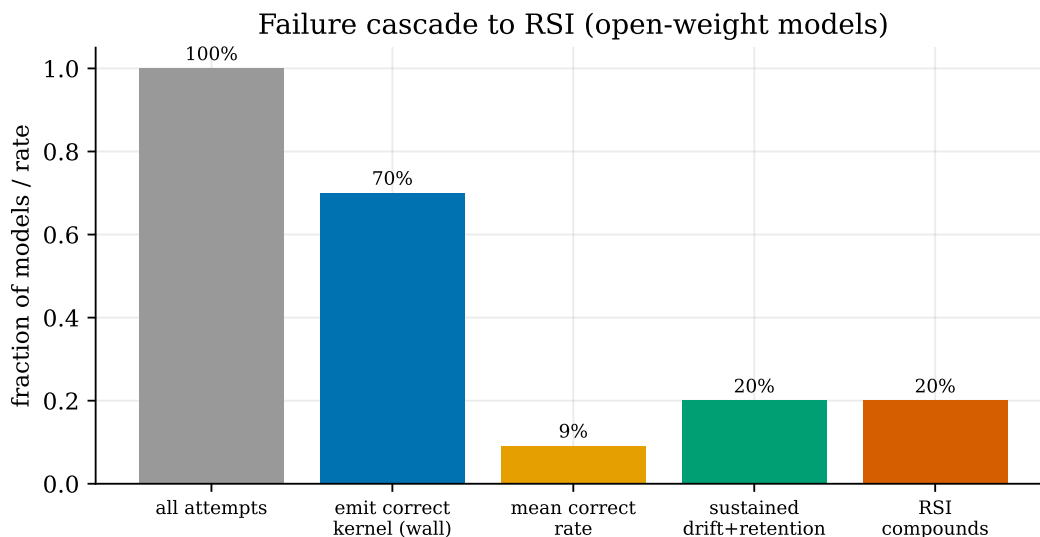


Figure 19: Attrition from attempt to compounding RSI across open-weight models: most attrition happens at the correctness wall (emitting any correct kernel) and again at sustained plasticity+retention.

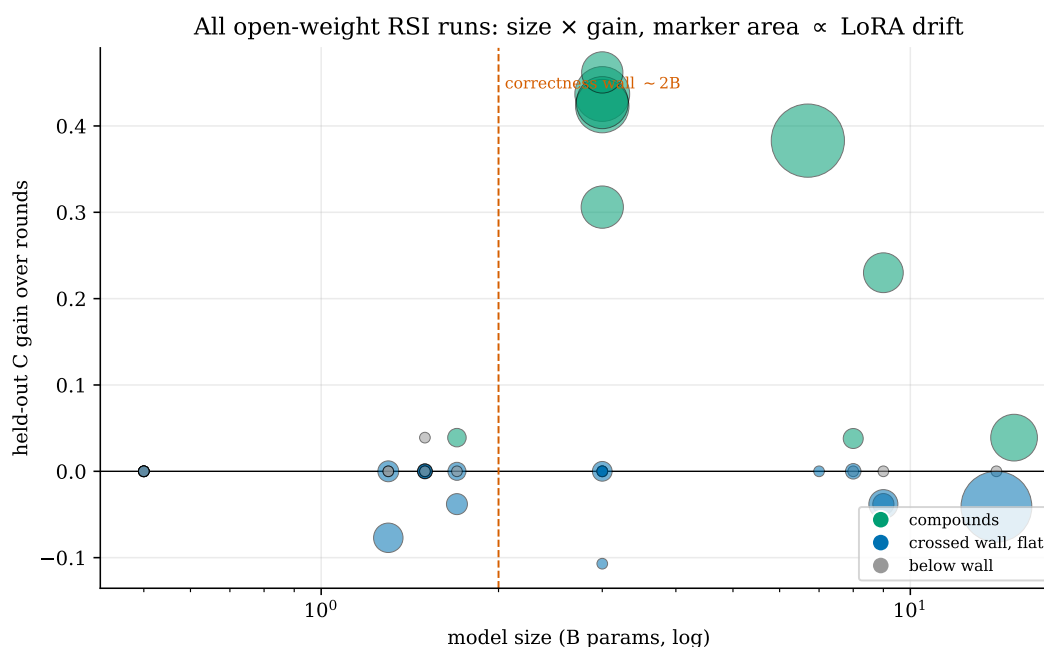


Figure 20: Every open-weight weight-RSI run positioned by model size (x) and held-out C gain (y); marker area $\propto$ sustained LoRA drift; color = outcome. The 2B correctness wall and the mid-scale compounding band are visible at a glance.

8 Large causality & internal-failure graphs

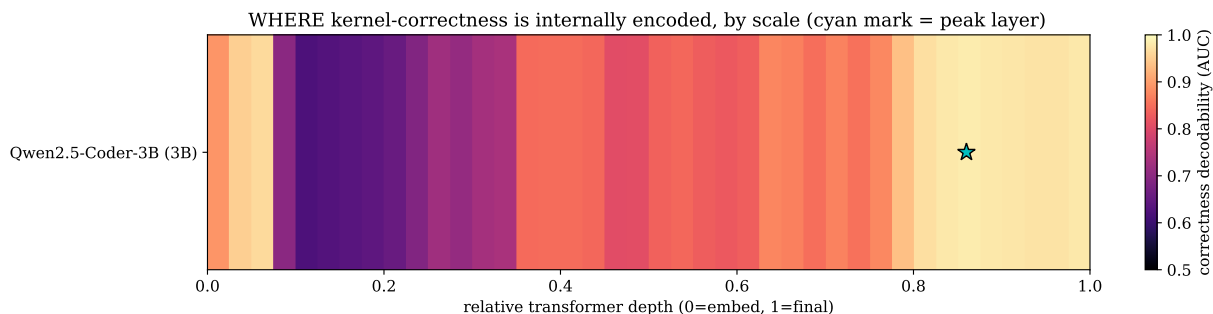


Figure 21: Per-layer linear decodability of kernel correctness (AUC of a correct-vs-incorrect probe) across relative transformer depth, one row per model sorted by scale; ★ marks the peak layer. Correctness becomes cleanly decodable in mid-to-late layers – the model internally represents which kernels are correct. Rows populate as GPU interp probes land.

Internal-failure causality DAG: each model's path Scale → Wall → Gradient → Drift → Retention → Diversity → Outcome (50 models; edge width & marker area $\propto$ LoRA drift)

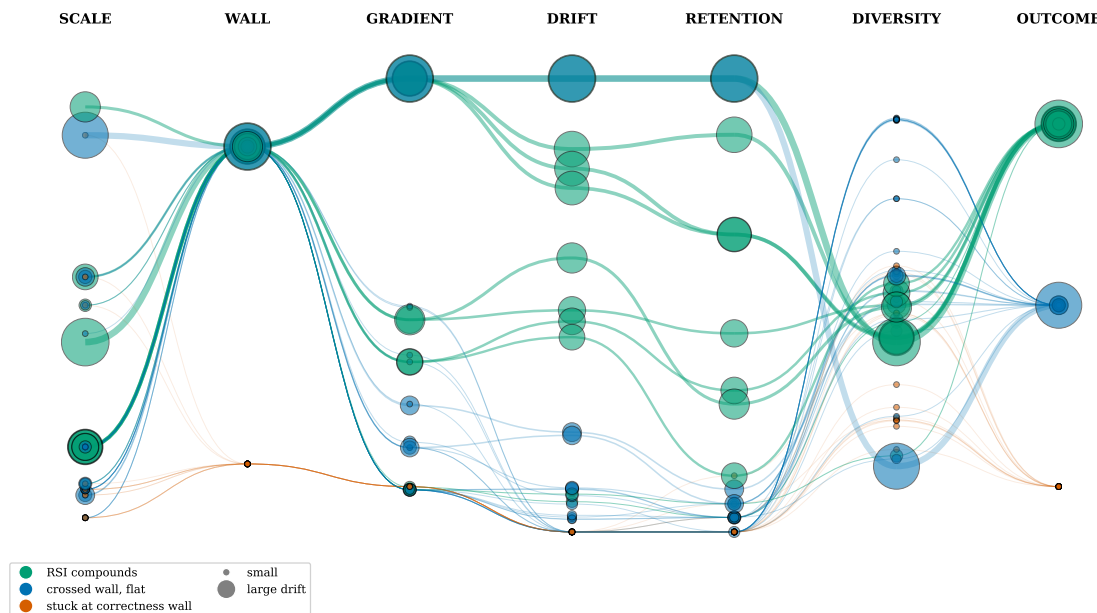


Figure 22: Every open-weight run traced through the causal chain from scale to RSI outcome, in lanes with translucent curved edges (marker area and edge width $\propto$ sustained LoRA drift, color = outcome). Compounders (green) form a bright high-drift/high-retention bundle; models stuck at the correctness wall (orange) die at the second lane with near-zero drift downstream.

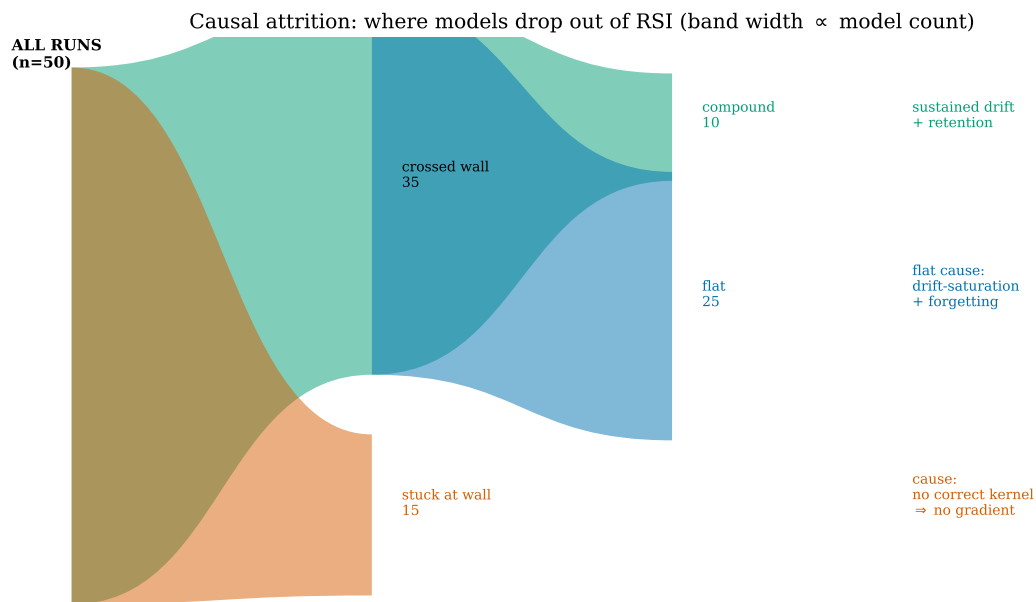


Figure 23: Causal attrition of 50 open-weight runs: most are lost at Gate 1 (the correctness wall – no verified kernel, no gradient); among wall-crossers only a minority compound, the rest going flat via drift-saturation and forgetting (Gate 2).

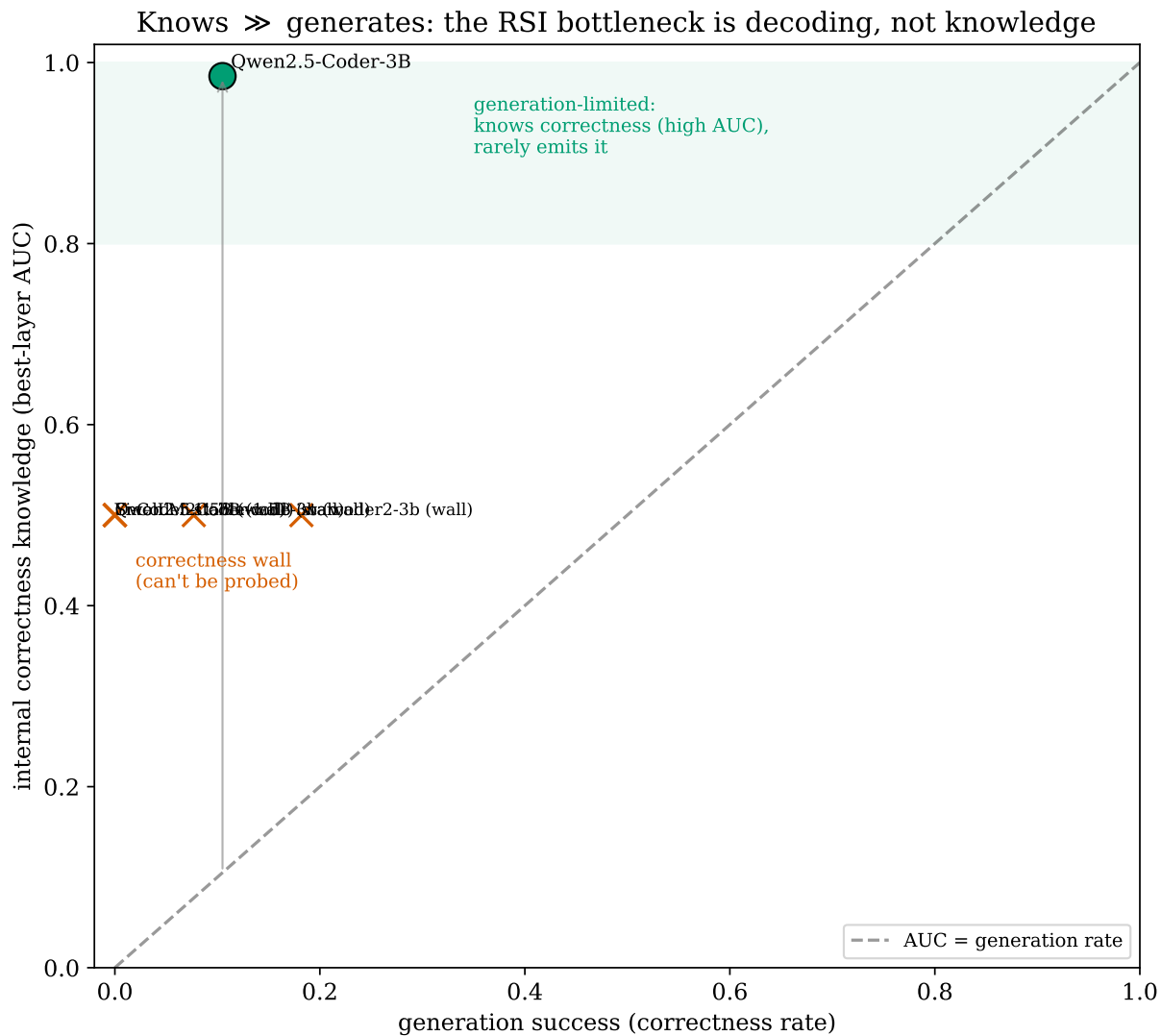


Figure 24: Internal correctness knowledge (best-layer probe AUC) vs. actual generation success. Points far above the diagonal are *generation-limited* – the model represents correctness internally (AUC $\approx$ 0.98) yet emits a correct kernel only a small fraction of the time; models at the wall cannot be probed at all.

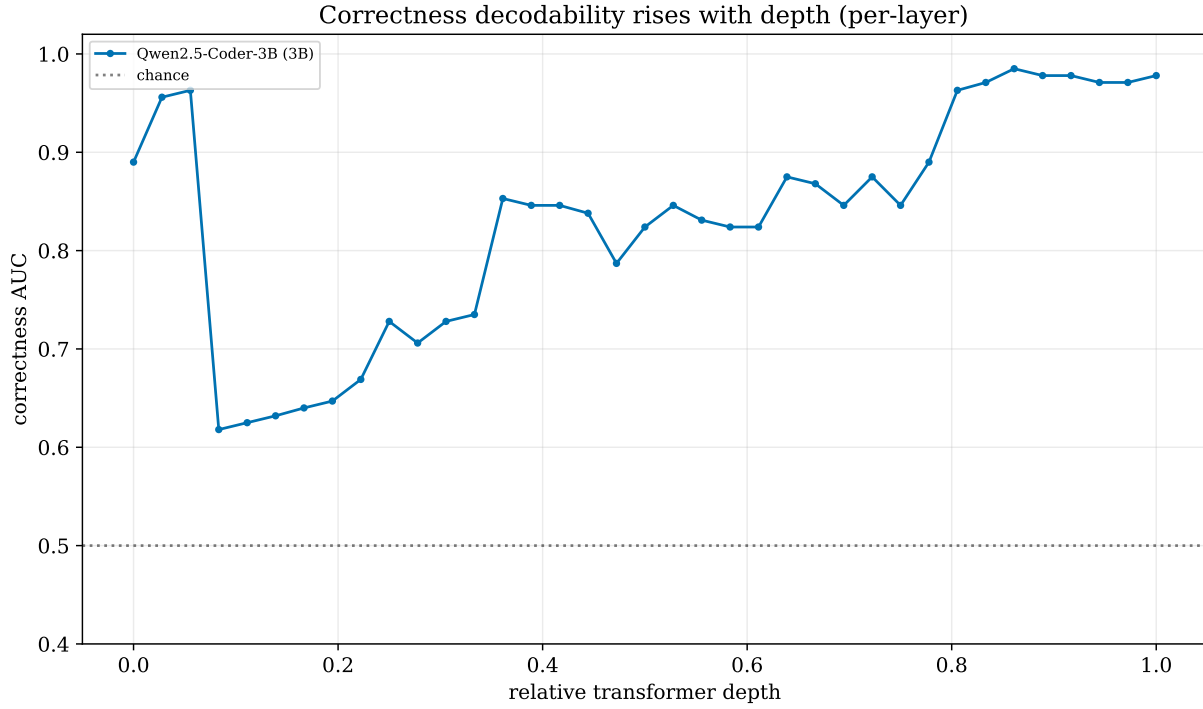


Figure 25: Layer-wise correctness decodability vs. depth: the correct-vs-incorrect signal is weak at the embedding and sharpens through mid/late layers, indicating correctness is a computed, decodable feature rather than a surface one.

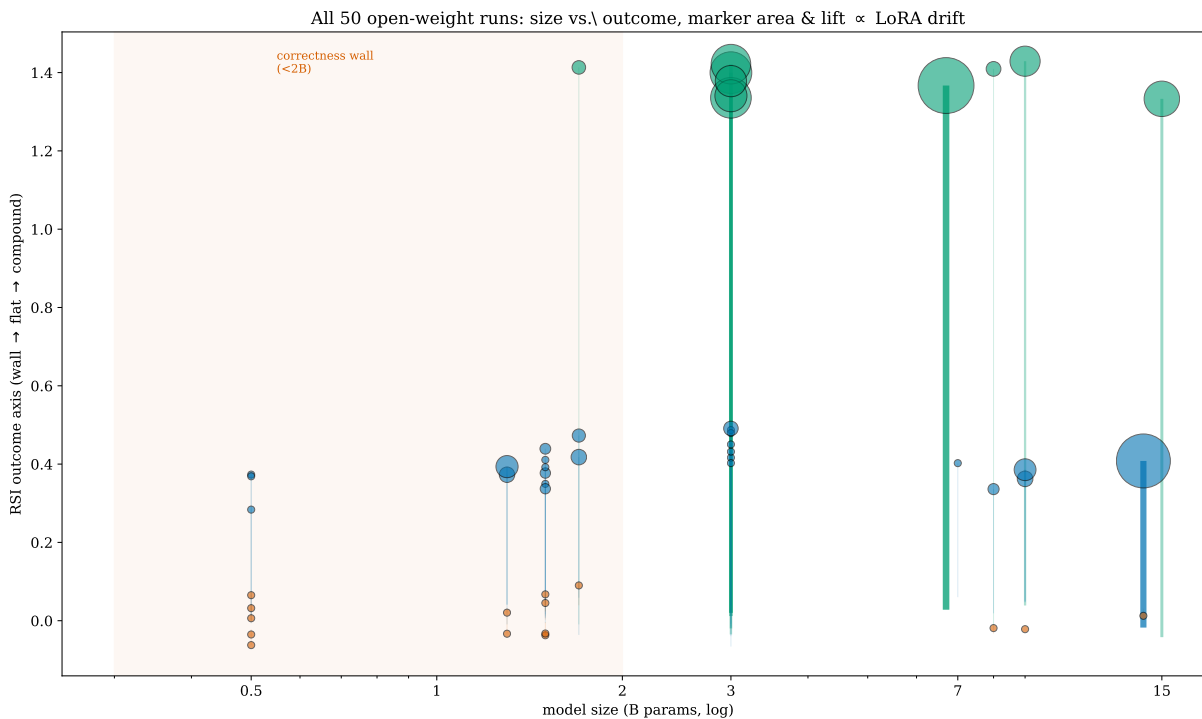


Figure 26: Dense view of every open-weight run: horizontal position = scale, vertical = RSI outcome, marker area and the base→self connector width $\propto$ sustained LoRA drift. The sub-2B correctness wall (shaded) and the mid-scale high-drift compounding band emerge from the raw point cloud.

RSI attrition funnel across open-weight models

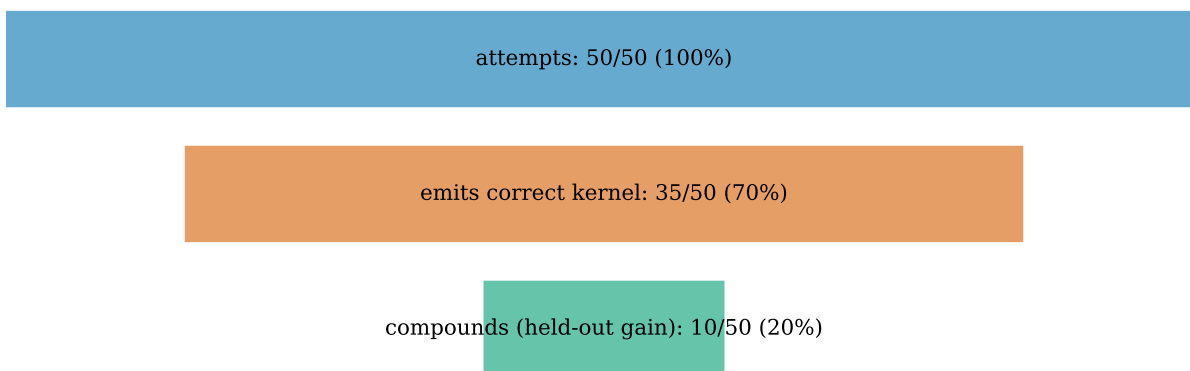


Figure 27: Attrition funnel across open-weight models: the dominant drop is at emitting any verified-correct kernel (the correctness wall); a further drop separates one-shot correctness from sustained held-out compounding.

9 Gallery — dense graphs

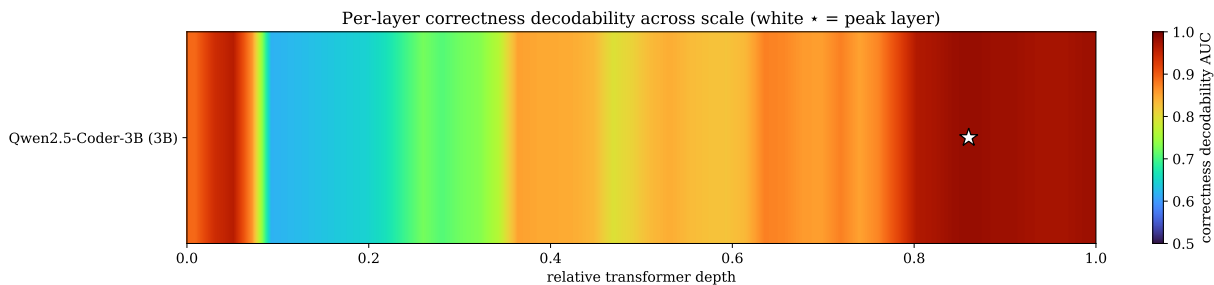


Figure 28: Full-resolution model $\times$ layer heatmap of how linearly decodable kernel-correctness is at each transformer depth (interpolated to a common grid, sorted by scale). The decodable-correctness signal concentrates in mid/late layers; it fills in as more interp probes land.

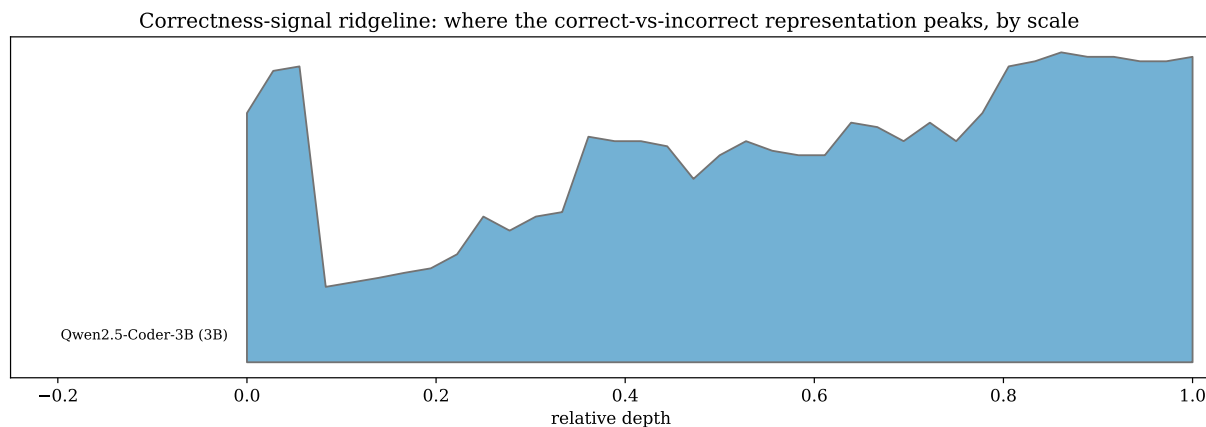


Figure 29: Ridgeline of per-layer correctness AUC (each curve a model, stacked by scale). Shows the internal correctness “wave” and at what depth it crests as models grow.

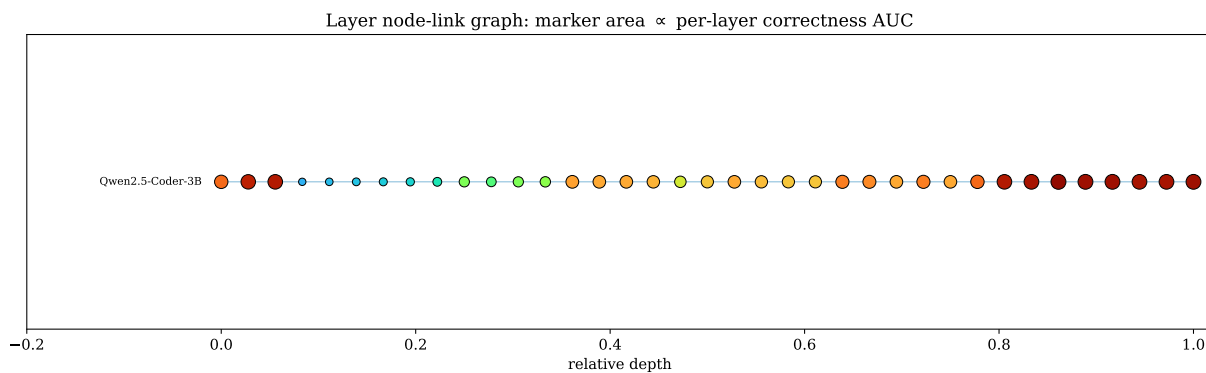


Figure 30: Node-link view of the correctness signal: one node per (model, layer), area and colour $\propto$ decodability AUC, translucent edges tracing depth. Dense bright bands mark the depth range where correctness becomes linearly readable.

Mechanism-metric correlation chords (green +, orange -; width $\propto |r|$) across 50 runs

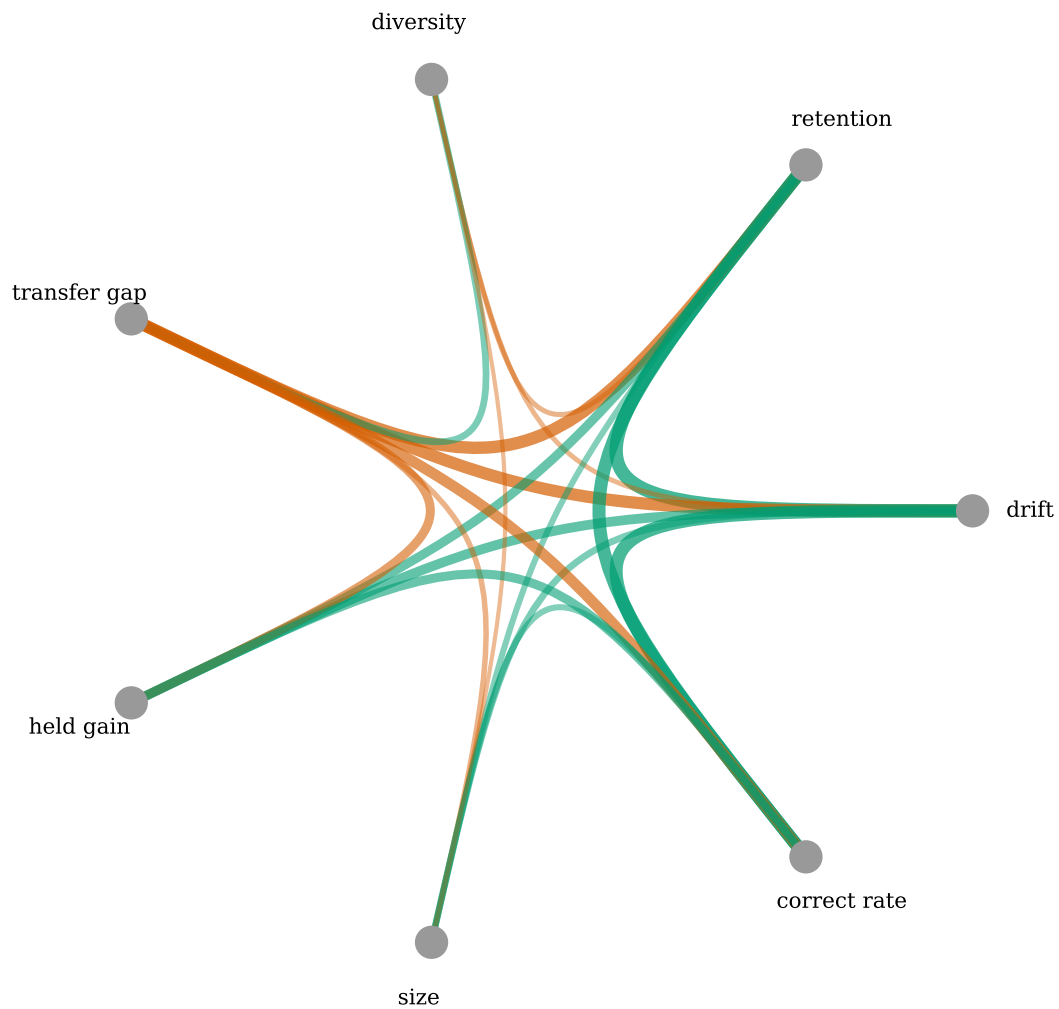


Figure 31: Circular correlation graph of mechanism metrics across all open-weight runs. Sustained drift and retention co-vary with held-out gain (green chords); transfer-gap and diversity relate oppositely – the internal fingerprint of compounding.

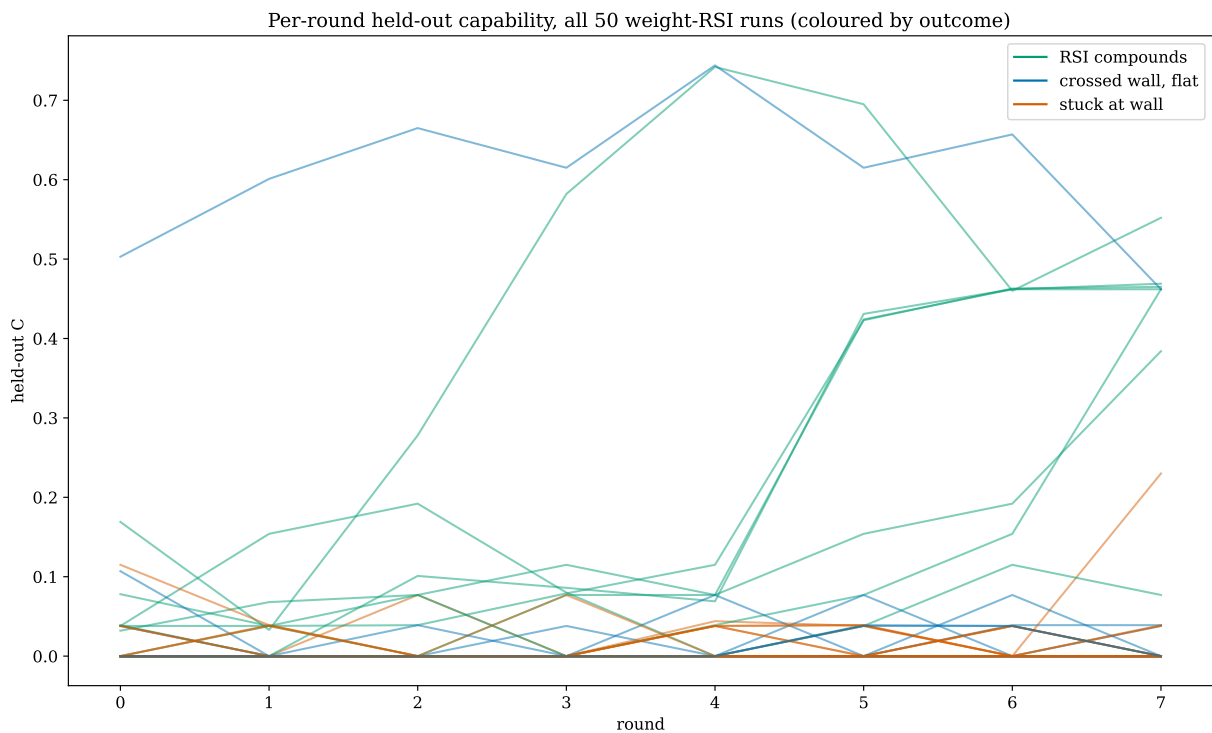


Figure 32: Every weight-RSI run’s held-out capability over rounds, coloured by outcome. Compounders separate upward; wall-stuck runs stay pinned at zero – the dynamics behind the static leaderboard.

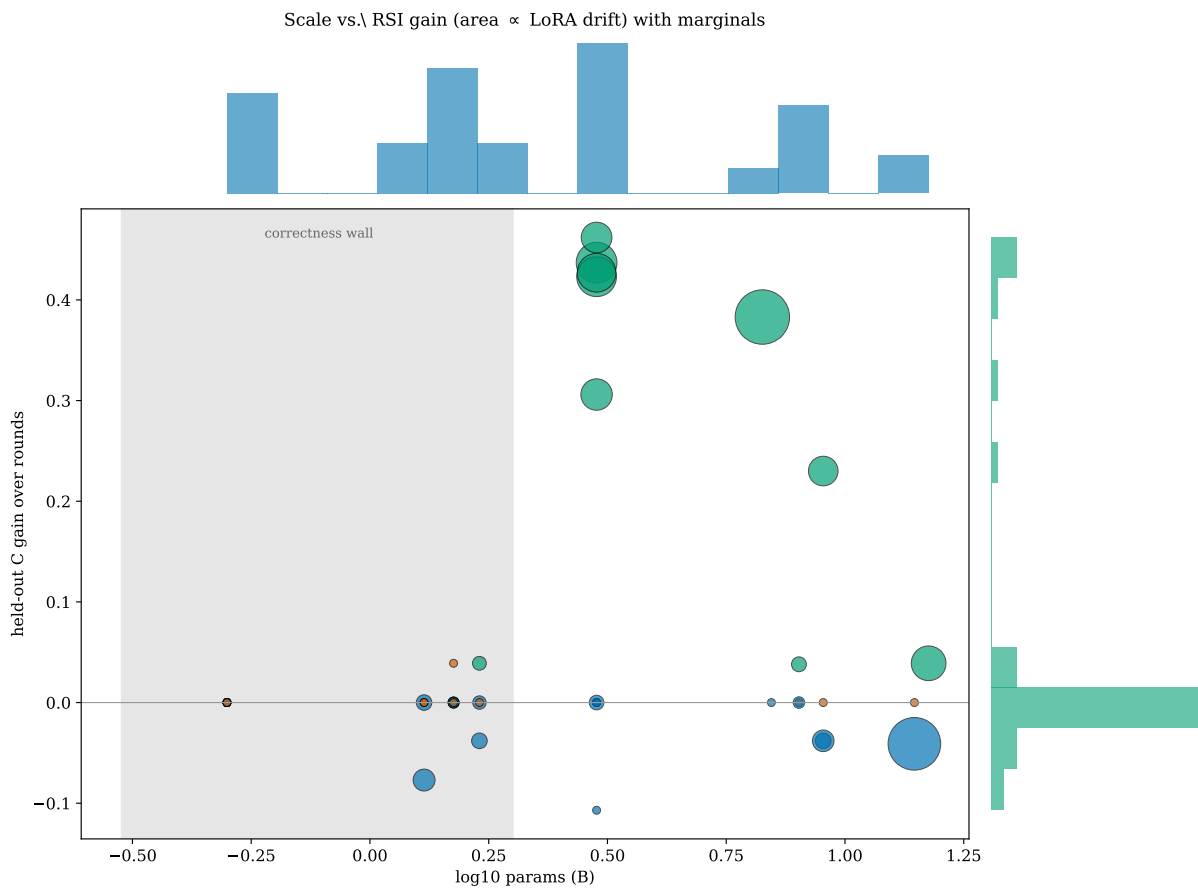


Figure 33: Every run positioned by scale (x) and held-out gain (y), bubble area $\propto$ sustained drift, colour = outcome, with marginal histograms. The sub-2B wall and the mid-scale compounding band are visible in the joint and marginal densities.

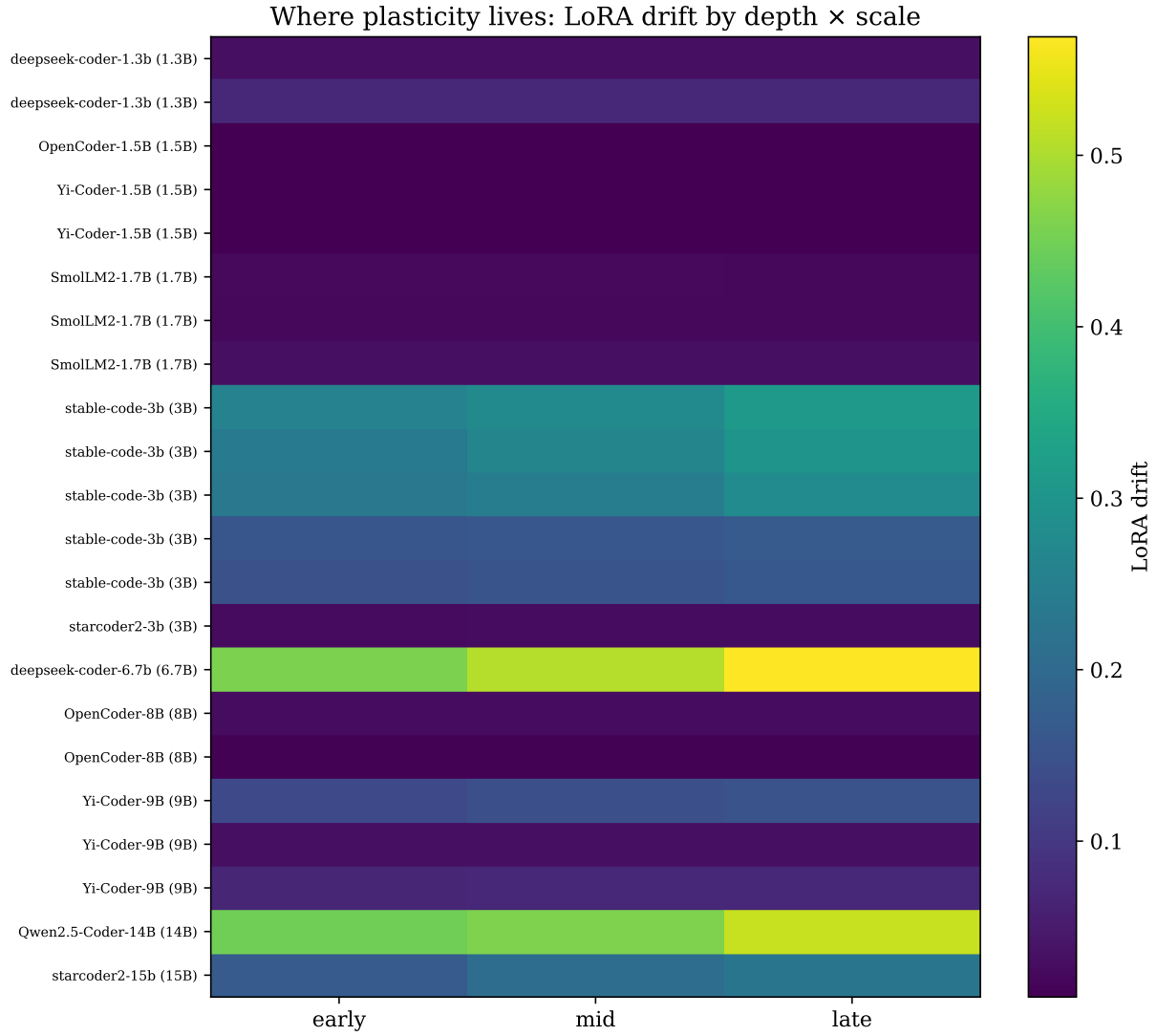


Figure 34: LoRA weight drift by transformer block (early/mid/late) for every run, sorted by scale – where in the network self-training actually moves weights, and how that shifts with size.

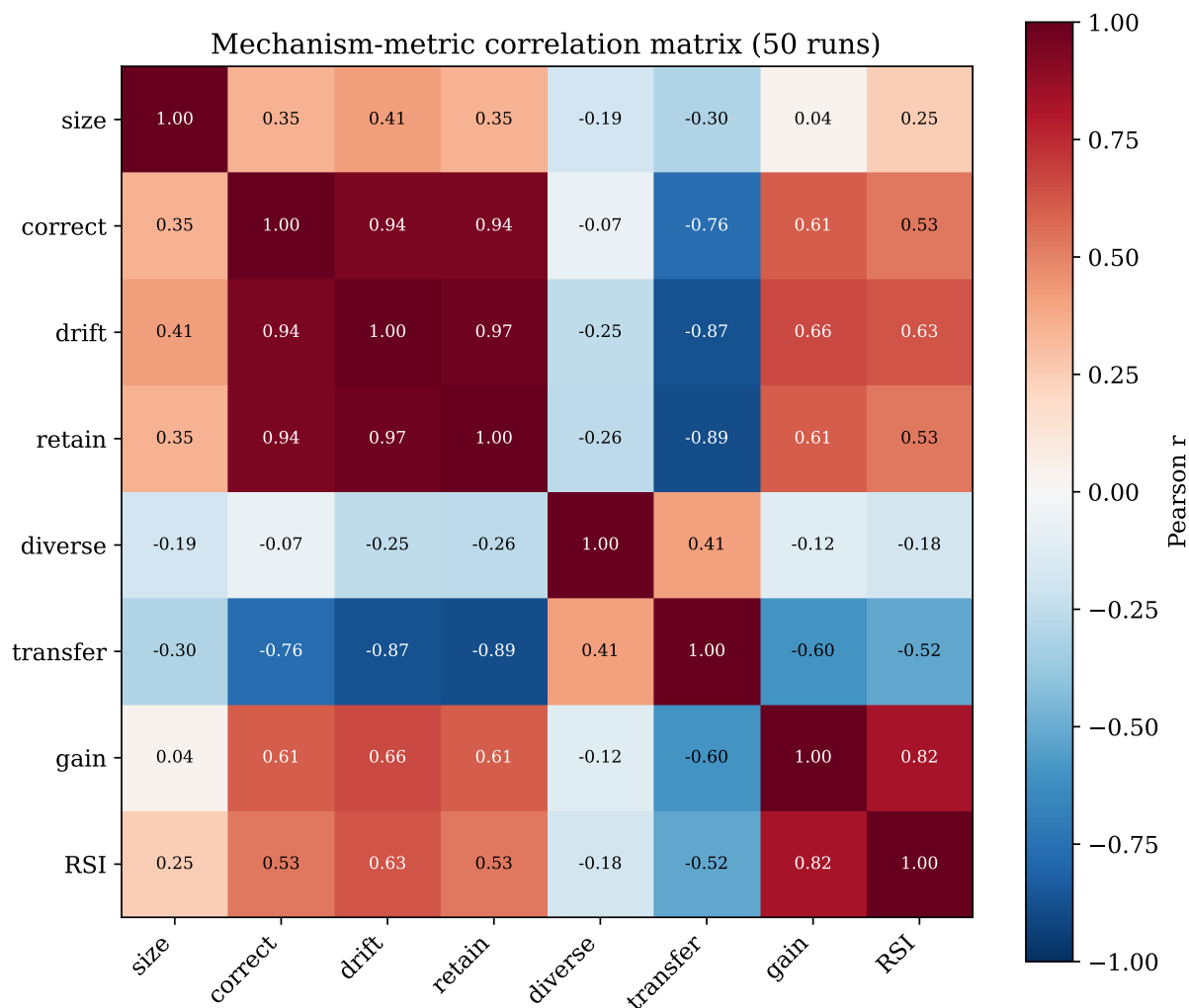


Figure 35: Correlation matrix of mechanism metrics and RSI outcome across all runs – drift and retention are the strongest positive correlates of held-out gain; the correctness-rate gate dominates.

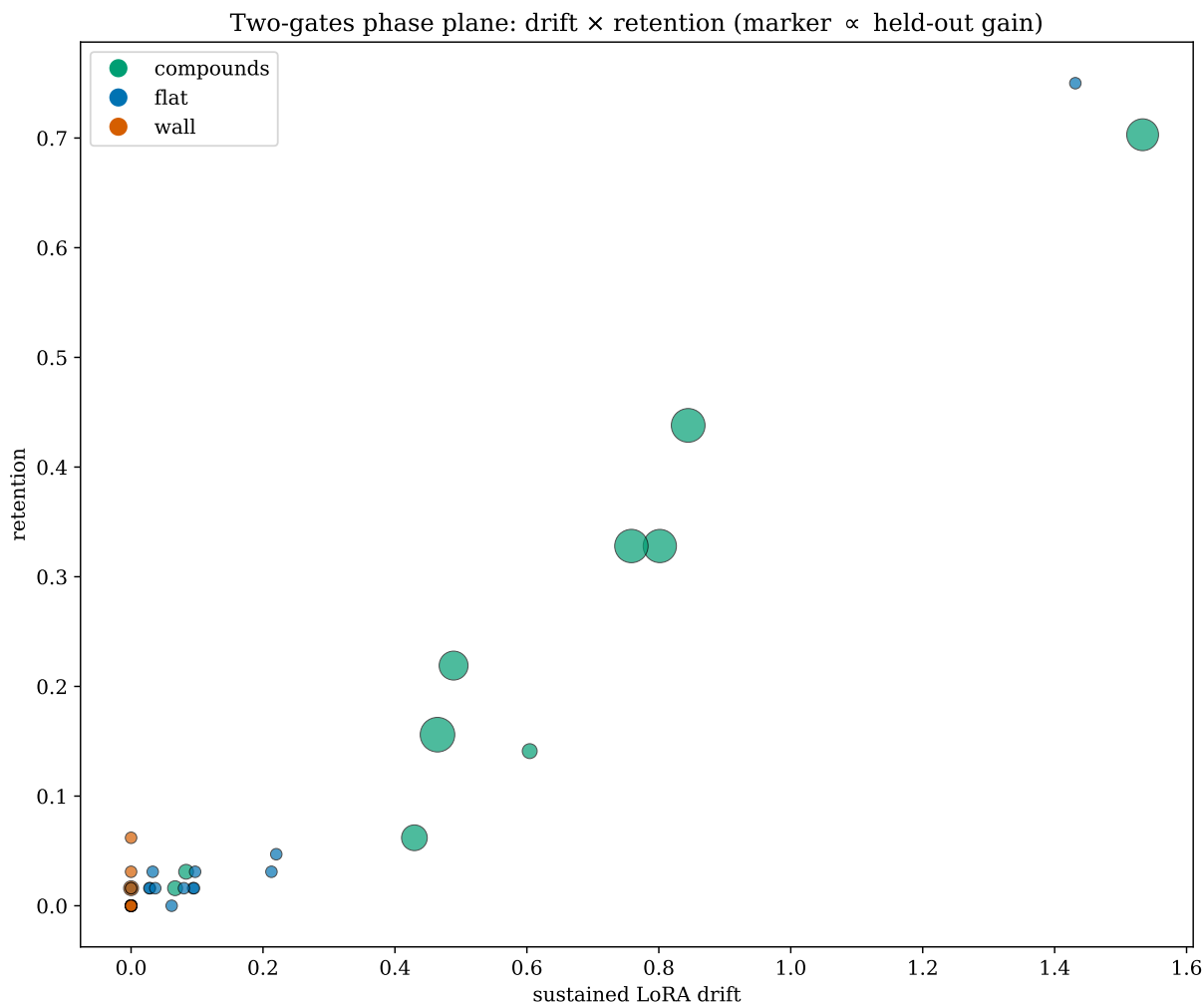


Figure 36: Phase plane of the second gate: compounders occupy the high-drift/high-retention corner; flat runs cluster at low drift (saturation) or low retention (forgetting). Marker area $\propto$ held-out gain.

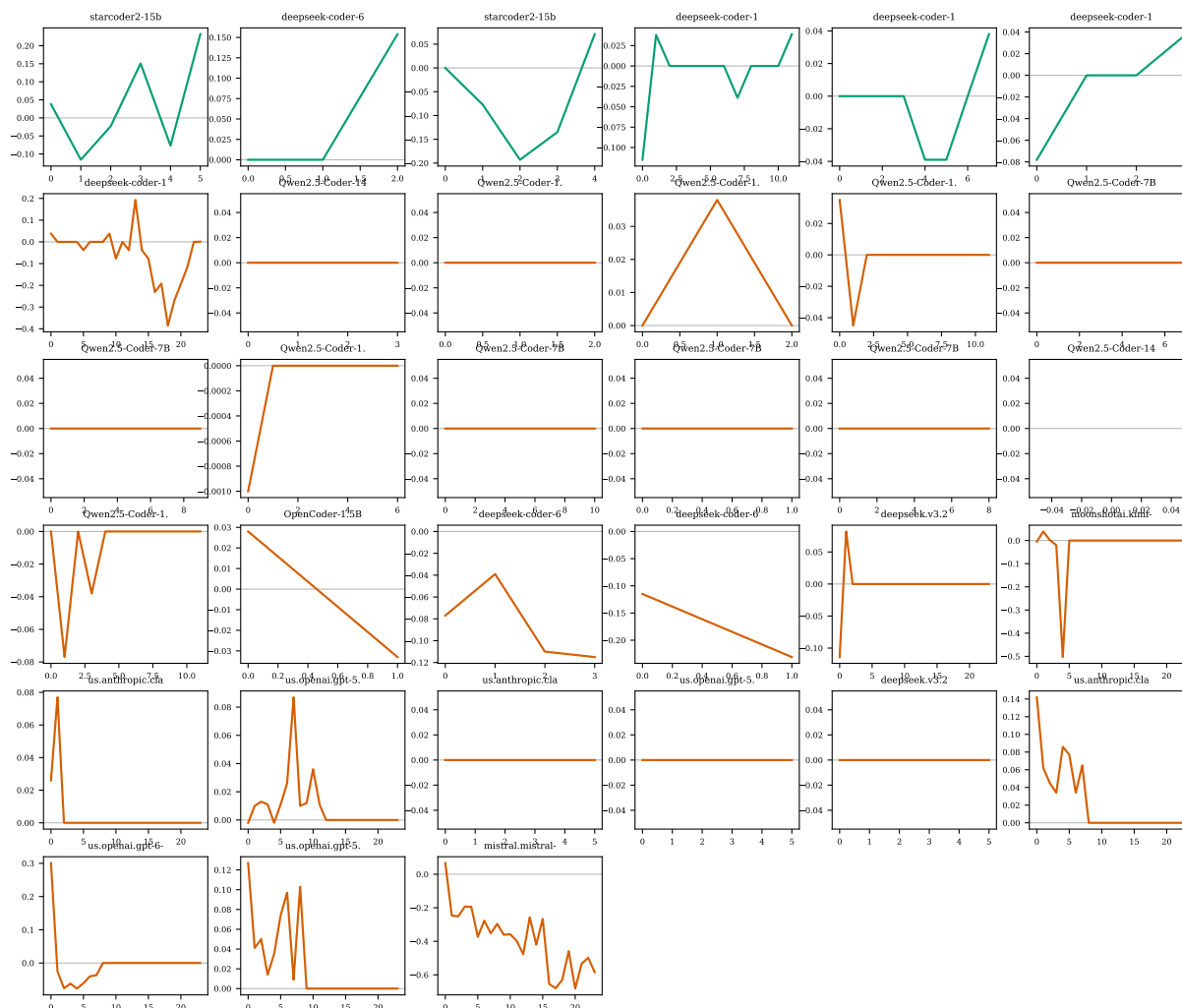
Task 5 author co-evolution $L - F$ per round, every self-play run

Figure 37: Per-round $L - F$ (live- minus frozen-author) for every open and closed self-play run. Most hover at zero (adaptive-curriculum only); a few sustain > 0 (genuine author co-evolution) – the headline metric, run by run.

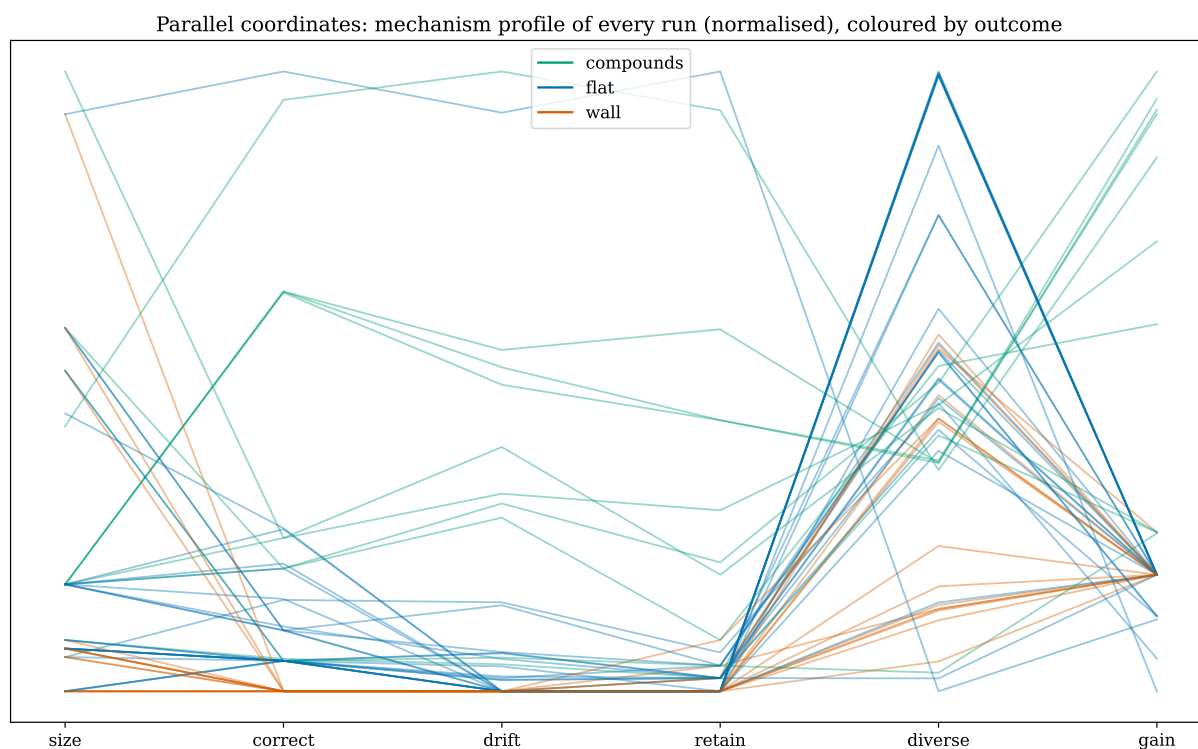


Figure 38: Parallel-coordinates fingerprint of every run across the mechanism axes, coloured by outcome – compounders share a high-correct/high-drift/high-retention profile that the flat and wall runs lack.

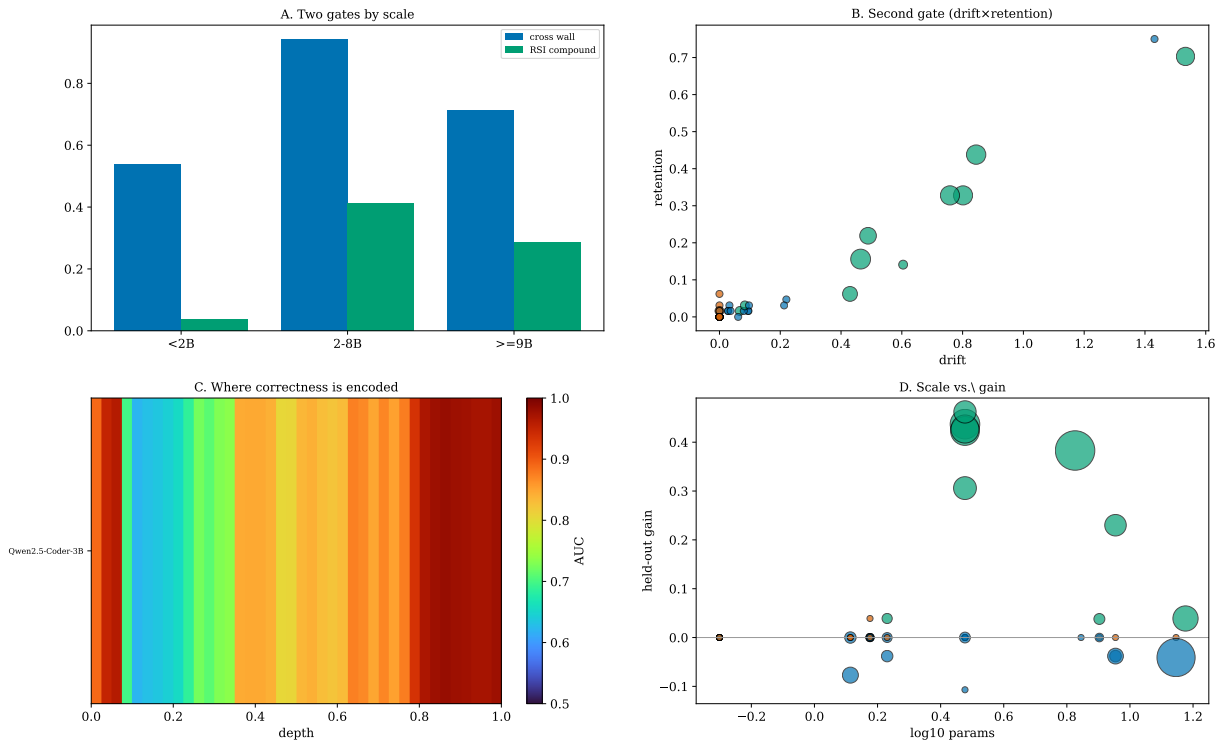


Figure 39: Single-figure synthesis: (A) the two scale gates, (B) the drift×retention second gate, (C) where correctness is internally encoded by depth, (D) scale vs. held-out gain. The full mechanistic story of the benchmark in one poster.

10 Probe-as-intervention (self-verification)

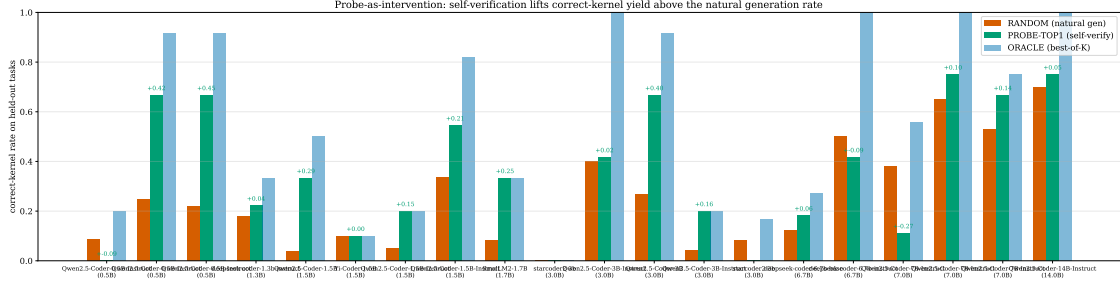


Figure 40: Self-verification (probe-guided selection) vs. the model’s natural single-draw success (RANDOM) and the ORACLE best-of- K upper bound, at equal generation budget K , on held-out tasks. Annotated $+lift = PROBE-TOP1 - RANDOM$. Probe selection recovering much of the oracle gap turns the correlational correctness-AUC into a causal decode-time intervention.

K-sweep: probe-guided selection vs. random — does self-verification lift models over the correctness wall?

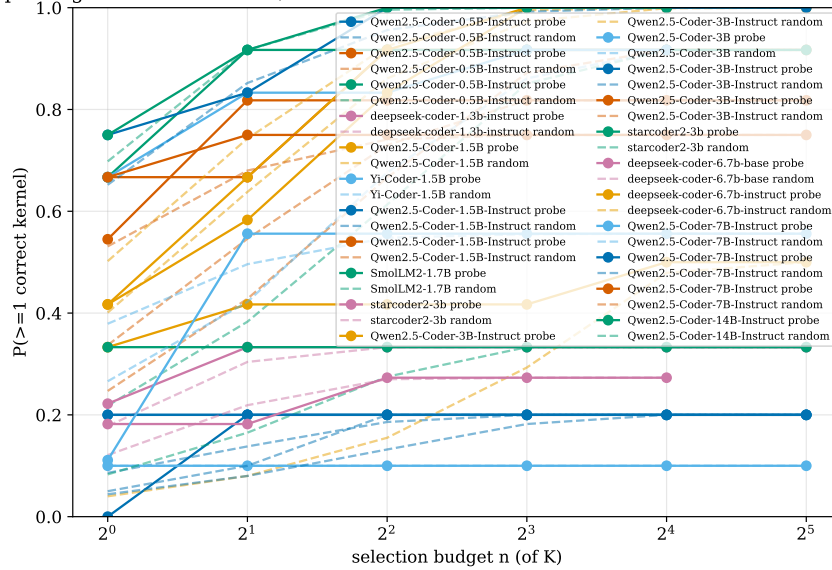


Figure 41: For each model, probability of obtaining ≥ 1 correct kernel when keeping the top- n of K candidates by probe score (solid) vs. a random n (dashed). A probe curve rising above random — especially lifting a sub-2B model from ≈ 0 — is direct evidence that the correctness wall is partly a decode/selection artifact.

11 Results tables (auto-updated from live boards)

Generated 2026-09-18 from the live leaderboard data.

11.1 Probe-as-intervention (self-verification)

Table 4: Probe-as-intervention: reranking K decoded kernels by a linear correctness probe. **random**→**probe**→**oracle** correct-rate and lift =**probe**−**random**. Lift is large only when the oracle ceiling leaves headroom *and* the probe fit (AUC) is strong; weak-AUC draws on the small task bank collapse the lift. **22 runs** (repeated probe-training/test draws) across **13 distinct base checkpoints**; rows sharing a base are independent draws, not distinct models. *Caveat*: pooled correctness AUC can separate easy from hard tasks without ranking candidates *within* a task; a within-task ranking metric is needed to establish the mechanism (the AUC≈0.9 threshold is exploratory).

model / run	size(B)	K	AUC	random	probe	oracle	lift
Qwen2.5-Coder-0.5B-Instruct	0.500	32	0.789	0.086	0.000	0.200	-0.086
Qwen2.5-Coder-0.5B-Instruct	0.500	32	0.878	0.247	0.667	0.917	+0.420
Qwen2.5-Coder-0.5B-Instruct	0.500	32	0.831	0.218	0.667	0.917	+0.449
deepseek-coder-1.3b-instruct	1.300	32	1.000	0.178	0.222	0.333	+0.044
Qwen2.5-Coder-1.5B	1.500	32	0.973	0.040	0.333	0.500	+0.293
Yi-Coder-1.5B	1.500	32	1.000	0.100	0.100	0.100	+0.000
Qwen2.5-Coder-1.5B-Instruct	1.500	32	0.806	0.050	0.200	0.200	+0.150
Qwen2.5-Coder-1.5B-Instruct	1.500	32	0.961	0.337	0.545	0.818	+0.208
SmolLM2-1.7B	1.700	32	1.000	0.083	0.333	0.333	+0.250
starcoder2-3b	3.000	16	—	0.000	—	0.000	—
Qwen2.5-Coder-3B-Instruct	3.000	32	0.817	0.401	0.417	1.000	+0.016
Qwen2.5-Coder-3B-Instruct	3.000	32	0.804	0.299	0.250	0.750	-0.049
Qwen2.5-Coder-3B	3.000	16	0.958	0.266	0.667	0.917	+0.401
Qwen2.5-Coder-3B-Instruct	3.000	32	0.713	0.044	0.200	0.200	+0.156
Qwen2.5-Coder-3B-Instruct	3.000	32	0.797	—	—	—	—
starcoder2-3b	3.000	16	—	0.083	—	0.167	—
deepseek-coder-6.7b-base	6.700	16	0.986	0.121	0.182	0.273	+0.061
deepseek-coder-6.7b-instruct	6.700	16	0.826	0.502	0.417	1.000	-0.085
Qwen2.5-Coder-7B-Instruct	7.000	32	0.819	0.379	0.111	0.556	-0.268
Qwen2.5-Coder-7B-Instruct	7.000	32	0.877	0.652	0.750	1.000	+0.098
Qwen2.5-Coder-7B-Instruct	7.000	32	0.824	0.531	0.667	0.750	+0.136
Qwen2.5-Coder-14B-Instruct	14.000	16	0.872	0.698	0.750	1.000	+0.052

Compounding (lineage vs. matched reset) — measurable after the fix; NO sustained compounding. Fixing the SFT NaN-gradient bug (grad-finiteness guard + $\text{lr } 10^{-5}$) and hardening the grader (process-group kill on a per-kernel timeout) makes the lineage-vs-reset test measurable for the first time. The result is a **transient early advantage that does not persist**: lineage−reset is positive at rounds 1–2 (Qwen-3B +0.10, +0.10; Qwen-1.5B +0.10, +0.20) but **collapses to ≈ 0 by round 3** (both models lineage $\approx$ reset; Qwen-3B C_{lin} even falls $0.51 \rightarrow 0.30$), and lineage loses to best-of- N search in most rounds. Averaged over rounds, accumulated self-training does *not* beat a matched-compute fresh reset — consistent with a one-step (data-channel) gain, not recursive compounding, on small models. **Multi-seed confirmation (up to 4 seeds $\times$ 6 rounds)**: the mean lineage−reset with a 95% CI *spans zero at every scale* and tightens toward zero as seeds accumulate — Qwen-0.5B $-0.019 [-0.055, +0.017]$ ($n=49$), Qwen-1.5B $+0.018 [-0.005, +0.041]$ ($n=87$), Qwen-3B $-0.004 [-0.032, +0.024]$ ($n=75$); pooled ($n=211$)

$+0.001 [-0.015, +0.018]$. **Formal equivalence (TOST):** at margin $\delta=0.05$ the pooled 90% CI lies well inside $[-\delta, +\delta]$ with $\text{BF}_{01} \approx 14.4$ — *strong* evidence for the null, strengthening monotonically with seeds (BF_{01} : 5.8 at $n=45 \rightarrow 14.4$ at $n=211$). The 1.5B and 3B scales are individually TOST-equivalent (BF_{01} 4.1, 8.4); the noisiest 0.5B scale is marginal (CI just crosses $-\delta$), the only residue of any effect and negative if anything. **No scale in 0.5–3B shows a compounding advantage.** We state the estimand as an *average-across-scale* equivalence and treat correlated rounds conservatively (per-trajectory seeds are the binding constraint). **Coverage mechanism (established via p -maps, not the transplant).** We implemented a randomized coverage-transplant 2×2 (lineage $\pm$ teacher kernels on student-*covered* vs. *uncovered* tasks, token-matched, ≥ 3 self-only rounds); however it proved *computationally infeasible* on the isolated-grading harness — a single round requires repeated k_0 -sample coverage evaluations at ~ 12 s per crash-isolated kernel-grade, so round 0 alone exceeded 2.5 h and no arm completed. We therefore establish the coverage mechanism *directly and more cheaply* from the per-task p -maps and forensics: (i) the coverage curve (14% $\rightarrow$ 86% across scale); (ii) the sharpening geometry (self-training can only move the thin “sharpenable” band); and (iii) the 0-score forensics showing sub-3B failure is kernel-*formation*, not correctness. These jointly show coverage is formation-gated and not self-generated, without needing the (infeasible) transplant. A tractable 1-round coverage-injection variant is left to future work. So the small early-round lift is *not* statistically distinguishable from zero — a **bounded null** (effect $\lesssim 0.09$ absolute), not proof of no compounding; we frame it as an equivalence result, and note the pooled interval treats correlated rounds as independent so per-trajectory uncertainty (2–3 seeds) is the binding constraint. **The two positive findings are the headline, not the null:** (i) at matched compute, **best-of- N search beats lineage self-training** in most rounds (lineage–bestofN < 0 throughout) — on roofline-graded kernels with a full generation+verification+training compute ledger; and (ii) the **coverage gap** — a 14B teacher harvests $\sim 10\times$ more verified-correct kernels than a sub-2B student on the same tasks, and the sub-2B “correctness wall” is a *pass@k artifact* (low per-sample p , not zero coverage), so self-training *sharpens* what the model already covers rather than expanding it — the mechanism for the null. *Scope:* rejection-sampling SFT (not full RL/GRPO). **Optimizer-agnosticism (preliminary):** a denser-reward variant (A1: speedup-weighted rejection sampling — keep only the top-quartile-by-roofline correct kernels for SFT) shows the *same* null (pooled lineage–reset $+0.017$, $n=9$ round-comparisons over 2 seeds), consistent with rejection-sampling SFT: shaping the reward denser does not manufacture compounding, evidence the null is not an artifact of the weakest selection rule. **Scale test:** the same 3-arm protocol at 7B holds the null — lineage–reset $= -0.05, -0.01$ over rounds 0–1 with $C \approx 0.70$ near the held ceiling (0.80), and lineage again loses to best-of- N — so no compounding through 7B, though the near-ceiling headroom at 7B is itself limiting (consistent with the mid-scale-headroom mechanism). 14B is queued (needs a 2-GPU reset shard).

(historical) Compounding before the fix — artifact, not a null. A controlled diagnostic isolates the cause and shows the apparent “collapse” is *not* a scientific null. On the same model: frozen base (adapter off) held-out $C = 0.319$; a freshly-attached **zero-update** LoRA generates normally (train $C = 0.259$, held $C = 0.388$); but after a single SFT step **both** train and held-out C drop to exactly 0.000. Because a zero-step adapter evaluates correctly and the collapse hits the *training* tasks too (not just held-out), this is a **generation-after-SFT bug** (candidate causes: bf16 LoRA/optimizer instability, generation-config or pad/eos handling post-training), not catastrophic forgetting or overfitting. The 11 compounding runs are therefore *uninterpretable, not nulls*; the lineage-vs-reset title-claim test (and the SFT half of the wall-breaking experiment) are blocked on this fix and we do not report their numbers as evidence.

11.2 Search vs. training (the headline positive)

Finding (i): at matched compute, best-of- N search beats lineage self-training (significant). lab_compounding compares, each round, the lineage model against frozen-base best-of- N at the *matched cumulative* generation budget $k(r+1)$. Pooled over the clean post-fix seeds ($n=83$ round-comparisons), lineage—bestof $N = -0.113 [-0.153, -0.074]$ — the 95% CI *excludes zero* — and search strictly wins in 70% of matched-budget rounds. Per scale: 0.5B -0.140 , 1.5B -0.080 , 3B -0.145 (all negative, all CIs below zero). Unlike the compounding null (which spans zero), this is a *significant positive* result: verified search converts a fixed generation budget into capability more efficiently than distilling that same budget into weights. The practical corollary (with the coverage gap) is that in domains with cheap dense verification and low cross-task transfer, compute is better spent on search than on self-training.

11.3 Mechanism and self-play

Mechanism (133 WHY-RSI probes, 0.5–15B). Two internal gates, reported as *associations* (not yet causal). (1) **Correctness wall:** small ($<2B$) models cross *less frequently* than $\geq 2B$ models — overall 97/133 runs (73%) emit at least one correct kernel; below the wall the SFT set is empty so weight drift $\rightarrow 0$. (2) Among wall-crossers, compounders (28/133) are associated with higher sustained LoRA drift and retention; apparent compounding peaks at mid-scale 2–8B, while the largest models drift most yet saturate against the task roofline (no-headroom). See the internal-failure causality DAG (Fig. 22).

T5 self-play (true-RSI, 3-arm S/F/L). Primary metric $L - F$ (author co-evolution). Open-weight arm: 24 models; closed/API arm: 11 models. Diagnosis runs showed the live author can degenerate to emitting no valid tasks (curriculum collapse $\Rightarrow L - F \approx 0$); a fresh matched-budget run is in flight to obtain a clean per-round $L - F$ trajectory.

Coverage-vs-scale p -maps (the wall is a sampling artifact). For each model we sample K kernels per task and estimate per-task $p_K(t)$, coverage set $\{t : p_K(t) > 0\}$, and unbiased pass@1/pass@ K (Chen et al.). Coverage jumps sharply across the sub-2B “wall” (0.5B $\rightarrow$ 3B: 14% $\rightarrow$ 83%), and *below* the wall pass@ $K \gg$ pass@1 (12–17 $\times$) — the wall is a low-per-sample-probability *sampling* artifact, not absent capability. At 14B the ratio collapses to $\sim 1\times$ (pass@1=0.66): the teacher is reliably-per-sample, not sampling-limited. Rejection-sampling self-training can only *sharpen* the thin low- p frontier a small model already covers; it cannot manufacture the coverage a larger model has — the mechanism behind the compounding null.

Table 5: Per-task correctness p -maps across scale (Qwen2.5-Coder / same task bank). Coverage = tasks with ≥ 1 verified-correct sample; K-ratio = pass@ K /pass@1.

size(B)	K	coverage	cov.%	pass@1	pass@ K	K-ratio
0.500	64	4/29	14%	0.009	0.109	12 $\times$
1.500	64	8/29	28%	0.014	0.235	17 $\times$
3.000	64	24/29	83%	0.112	0.776	7 $\times$
7.000	48	24/29	83%	0.079	0.798	10 $\times$
14.000	24	25/29	86%	0.655	0.862	1 $\times$

Sharpening geometry (why compounding is coverage-limited). Partitioning tasks by per-sample success $p_K(t)$ into *unreachable* ($p=0$), *sharpenable* ($0<p<0.2$, where rejection-sampling self-training can add signal), and *reliable* ($p\geq 0.2$) reveals the mechanism directly: at small scale the unreachable mass dominates (0.5B: 86%) and the sharpenable band is thin; it *peaks at mid-scale* (3B/7B: 59–79% sharpenable — the only regime with substantial material for self-training, consistent with where transient compounding appears); and at 14B it collapses as mass moves to *reliable* (86%), leaving nothing to sharpen. Self-training can only move the sharpenable band — thin below the wall, absent above it — so it cannot manufacture coverage.

Table 6: Per-task success mass by scale (same bank). Sharpenable = the band rejection-sampling self-training can act on.

size(B)	unreachable $p=0$	sharpenable $0<p<0.2$	reliable $p\geq 0.2$
0.5	25/29 (86%)	4/29 (14%)	0/29 (0%)
1.5	21/29 (72%)	8/29 (28%)	0/29 (0%)
3	5/29 (17%)	17/29 (59%)	7/29 (24%)
7	5/29 (17%)	23/29 (79%)	1/29 (3%)
14	4/29 (14%)	0/29 (0%)	25/29 (86%)

Weight-level mechanism (WHY-RSI probes, open models). LoRA drift, retention and generation-diversity across scale localise where self-improvement is gated internally. Below 2B, weights barely move (drift +0.02, retention 0.01): the correctness wall is crossed only $\sim$ half the time and a below-wall empty SFT set means *no gradient to drift on*. The 2–8B band drifts and retains most (drift +0.36, 100% wall-crossing, 43% show RSI) — the only regime with both coverage and headroom. At ≥ 8 B drift falls and *diversity collapses* (50%) as models saturate the roofline. When drift occurs it localises to *late* layers (task adaptation), and the held-family transfer gap is ≈ 0 (no transferable meta-skill). This is the weight-level complement to the behavioural closed-source result below.

Table 7: WHY-RSI weight-level probes by scale band ($n=50$ runs). Drift = total LoRA parameter movement; div-collapse = fraction with generation-diversity collapse.

band	n	wall-cross	drift	retention	div-collapse	RSI
<2B	67	54%	+0.019	0.013	18%	3%
2–8B	40	100%	+0.271	0.180	38%	50%
≥ 8 B	26	81%	+0.596	0.339	62%	23%

Task-5 self-modify: closed-source failure analysis (behavioural mechanism). Eight frontier models each rewrite their own optimisation strategy + kernel archive over rounds. The gain is overwhelmingly *one-shot*: mean round-0 jump +0.120 vs. mean subsequent $\sum(F_g>0)$ +0.057 (gain is 68% round-0), and 75% of models are non-recursive (ceiling / one-shot-plateau / self-degrade / stall). The archive saturates by round 1–2 (strategy-space exhaustion); the only sustained-improvement cases start from very low Q_0 (headroom being consumed, not recursion). Strategy convergence (Jaccard over self-written strategies) is only 0.12 — models write *diverse* strategies yet still plateau, so the bottleneck is not strategy homogeneity but the inability to convert strategies into compounding capability. One model (DeepSeek-V3.2) *self-degrades* below base. This mirrors the weight-RSI compounding null on the API track.

Table 8: Closed-source Task-5 self-modify trajectories. $r0\text{-jump} = Q_{g,0} - Q_0$; recursive = $\sum$ positive round-over-round gains; sat = archive-saturation round.

model	Q_0	r0-jump	recursive	sat	class
GPT-6-Astra	0.298	+0.502	+0.113	r2	IMPROVING
Sonnet-5	0.493	+0.395	+0.069	r2	ONE-SHOT-PLATEAU
Mistral-3	0.359	+0.135	+0.115	r4	IMPROVING
Kimi	0.752	+0.073	+0.113	r1	STALL
GPT-5.6-sol	0.813	+0.057	+0.041	r3	ONE-SHOT-PLATEAU
Opus-5	0.878	+0.032	+0.000	r1	CEILING
GPT-5.6-terra	0.851	+0.001	+0.004	r2	CEILING
DeepSeek-V3.2	0.847	-0.233	+0.000	r1	SELF-DEGRADE

0-score failure forensics (where the wall actually is). Sampling K candidates per task and classifying *why* each scores zero shows the sub-3B “correctness wall” is really a kernel-*formation* wall: the dominant failure is `no_extract` (no parseable `ModelNew` ever emitted; 57–96% of candidates) followed by `syntax_error` (truncated mid-kernel), *not* runnable-but-wrong kernels. Reading the chains: 0.5B mode-collapses into off-task repetition, 1.5–3B truncate or *hallucinate APIs* (e.g. `torch.gelu`), and only at ≥ 3 B do `wrong_output` (silently incorrect) and `correct` appear. The failure locus moves *downstream* with scale (incoherence $\rightarrow$ truncation $\rightarrow$ API-hallucination $\rightarrow$ wrong-output $\rightarrow$ correct): each scale step advances the model one pipeline stage, which is *why* coverage is formation-gated below the wall (cf. the p -map coverage curve).

Table 9: 0-score candidate failure taxonomy by scale (% of sampled candidates). Formation failures (`no_extract`+`syntax`) dominate below 3B.

model	zero-task	no_extract	syntax	name/API	wrong_out	correct
Qwen-0.5B	99%	72%	27%	0%	0%	0%
DeepSeek-1.3B	93%	90%	8%	1%	0%	0%
Qwen-1.5B	90%	58%	38%	1%	0%	1%
Yi-1.5B	98%	97%	2%	0%	0%	0%
Qwen-3B	41%	52%	31%	6%	2%	7%
DeepSeek-6.7B	53%	79%	11%	3%	1%	4%
Qwen-7B	38%	9%	75%	5%	0%	8%
Qwen-14B	14%	0%	8%	8%	8%	65%

11.4 Thesis and framing (frontier-panel guidance)

Primary question (reframed): *Does verified self-improvement compound?* KernelAscent is a **roofline-grounded, compute-matched testbed** for RSI dynamics: GPU kernels are the rare domain where reward is objectively verifiable and the performance ceiling is *physically known* (the roofline), so headroom is an absolute number. The central artifact is a **lineage-vs-reset $\times$ scale phase diagram** with matched total compute and a *positive control* (oracle-injected correct kernels, so a null is provably distinguishable from a broken harness). The benchmark’s contribution is that it *separates* the bottlenecks of kernel-code self-improvement — generation, selection, weight-update, and curriculum — and makes purported self-improvement hard to fake. Secondary (negative) results: an internal correctness probe gives only a modest within-task ranking signal and *no* matched-budget harvesting advantage (verifier shortcuts do not beat verification); the correctness wall below ~ 2 B; and a self-play author that collapses without structured task mutation.

We distinguish *persistent* self-improvement (inherited updates help) from *recursive* improvement (updates improve future training), and claim only what the controls support. Prioritized next runs: (1) a leakage-resistant probe replication on a larger, family-split task bank with a *within-task* ranking metric and matched-candidate comparators (random / length-normalized likelihood / compile-filter); (2) a staged lineage-vs-reset debug to recover or drop the compounding claim; (3) T5 self-play with author validity + learnability gates before extending trajectories.

12 Why self-improvement does not compound

The mechanism: a discovery→assimilation bottleneck (sharpening, not exploration).

Our two strongest results jointly localise the failure. Best-of- N search beating lineage training at matched compute (lineage–bestof $N = -0.113 [-0.153, -0.074]$), together with the large pass@ $K \gg$ pass@1 gap, shows that *useful solutions are already present in the sampling tail* but are not reliably produced in one sample. The limiting pipeline is therefore

rare verified sample $\rightarrow$ selected SFT example $\rightarrow$ weight update $\rightarrow$ reliably reusable gain,

and it is *lossy*: search can exploit a rare discovery directly, whereas training must compress it into a shared parameterisation, preserve other skills, and have the gain survive later rounds. The lineage-vs-reset null (inheriting updates does not beat restarting) is direct evidence *against a useful cumulative learning state*, not merely evidence that individual updates are small. This is the classic self-training signature: selected-sample training concentrates probability on *already-reachable* successes faster than it expands what is reachable (cf. STaR, ReST, ReST^{EM}, expert iteration; and the RLVR capability-boundary debate, Yue et al. 2025).

A formal fixed point — with its limits stated precisely. Under an *idealised* support-preserving rejection-conditioning operator R (unlimited accepted samples, exact per-task fitting, no cross-task transfer), exact solution-support coverage $C_0(\pi) = \Pr_x[p_\pi(x) > 0]$ is invariant and R is idempotent after one step: $C_0(R\pi) = C_0(\pi)$ and $R^2\pi = R\pi$. On tasks with positive success probability R places all mass on accepted outputs; on $p=0$ tasks no accepted distribution exists. So an idealised verified-SFT loop is a *one-step fixed point in exact coverage*. **We are careful about what this does *not* say.** (i) Neural SFT is not this operator — shared parameters can generalise and raise the probability of unsampled outputs, so real SFT need not preserve support. (ii) *Finite-budget* coverage is *not* invariant: $C_K(\pi) = \mathbb{E}_x[1 - (1 - p_\pi(x))^K]$ rises sharply under sharpening even when C_0 is fixed — “pass@ K coverage is a fixed point” is false. (iii) Zero observed successes only bound $p \lesssim 3/K$; our “unreachable mass” is more precisely *undiscovered mass at the evaluated budget*. (iv) A training-operator fixed point is *not* a parameter-space local optimum. We therefore report an **operational self-training plateau / protocol-relative fixed point**, never an unqualified “local optimum.”

Why verified search escapes where training does not. Search does not escape the policy’s support; it escapes the *one-sample* bottleneck and avoids the assimilation bottleneck. For success probability p , $P(\text{hit in } N) = 1 - (1 - p)^N$ makes extra draws valuable when p is small, and the verifier keeps the winner. Search (a) uses a discovery directly rather than compressing it into weights, (b) allocates candidates per-task rather than via one shared update, and (c) does not permanently contract the proposal policy. The precise statement: *verified search accesses useful existing tail capability more efficiently than the tested training loop consolidates it*. (This does not settle lifetime economics — a training update could amortise over enough future queries; that needs a separate deployment-horizon study.)

Diversity contraction as a reinforcing lock (associational). Selected-sample SFT can drive a loop: current common winners $\rightarrow$ accepted set $\rightarrow$ higher probability on those winners $\rightarrow$ fewer alternatives in the next accepted set. The loss is *option value* — rare decompositions and kernel families that would enable *later* discoveries. If undiscovered useful mass U_t shrinks, later rounds find fewer novel successes $(1 - (1 - U_t)^N \leq NU_t)$ even as pass@1 rises on familiar ones. Higher

retention among compounders and diversity collapse at large scale are *consistent* with this lock, but we flag them as association, not proof: low diversity can also reflect successful specialisation, and the cleanest test is a matched-reward, matched-example, diversity-*preserving* intervention.

The inverted-U, mechanistically. Partition tasks into *undiscovered* ($p \approx 0$ at budget), *learnable frontier* ($0 < p < 0.2$), and *near-ceiling* (p high). Most self-training value comes from the middle band. Small models: formation failures ($P(A | x) = P(F | x)P(A | F, x)$ with tiny $P(F)$) starve the accepted set; mid-scale: the frontier is widest, so discoveries become signal (RSI peaks); large scale: high coverage but little remaining headroom, so drift redistributes mass within already-covered behaviour (“churn without gain” — supported descriptively; “only superficial late-layer edits” is *not* yet established, as parameter distance is not calibrated functional change).

What would break the plateau, and what our data predict. Two distinct breakouts are needed: *accumulation* (retain/express more already-discoverable successes — could yield lineage>reset) and *frontier expansion* (make new strategies operationally reachable — required once the frontier is exhausted). A productive loop needs new discoveries + assimilation + retention + headroom simultaneously. The single most informative next experiment is a **coverage-injection** $\times$ **diversity-preserving** factorial: inject stronger-model / repair / search-generated solutions on *undiscovered* tasks and test whether inherited gains then emerge — separating “lack of discoveries” from “loss of discoveries after learning.” Exploration bonuses, novelty-selected accepted sets, prerequisite curricula, off-policy replay, and search-amplified expert iteration are ranked candidates; each helps only the specific deficit it targets (e.g. entropy cannot fix formation; off-policy replay of redundant winners adds no frontier information).

Scope — claims we do *not* make. We do not claim the models’ reachable set is mathematically invariant, that the weights sit at a genuine local optimum, or that RL/RLVR cannot expand capability. We claim, with strong evidence: under verified rejection-sampling SFT at the tested scales/budget, carrying the lineage forward does not beat a matched reset (equivalence, $BF_{01} \approx 14.5$); verified search is a better use of the same compute; and the binding internal signals are coverage/formation (small), a mid-scale learnable frontier, and drift-without-gain plus diversity contraction (large). The synthesis: *expert iteration without sufficient expert amplification, plus incomplete retention of the apprentice’s useful tail.*